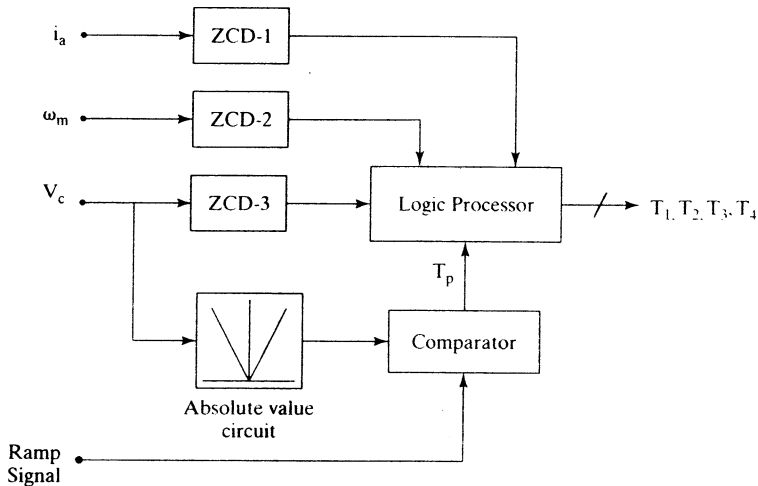


This logic amounts to the fact that the duration for which the control signal exceeds the ramp signal determines the duty cycle of the chopper. The on- and off-time signals are combined with other control features, such as interlock, minimum on- and off-times, and quadrant selection. The interlock feature prevents the turning on of the transistor (top/bottom) in the same leg before the other transistor (bottom/top) is turned off completely. This is ensured by giving a time delay between the turn-off instant of one device and the turn-on instant of the other device in the same phase leg. Simultaneous conduction of the top and bottom devices in the same leg results in a short circuit of the dc source; it is known as shoot-through failure in the literature.

Figure 4.21 corresponds to the forward motoring operation in the four-quadrant chopper shown in Figure 4.2. When the motor drive is to operate in the third and fourth quadrants, the armature current reverses. This calls for a change in the current-control circuitry. A block diagram of the current controller is shown in Figure 4.22, including all the constraints pertaining to the operational quadrant. The speed and current polarities, along with that of the control voltage, determine the quadrant and hence the appropriate gating signals. The on-time is determined by comparing the ramp signal with the absolute value of the control voltage. The current-error signal, which determines the control voltage v_c , is rectified to find the intersection point between the carrier ramp and v_c . A unidirectional carrier-ramp signal can be used when the control voltage is also unidirectional, but the control voltage will be negative when the current error becomes negative. It happens for various cases, such as reducing the reference during transient operation and changing the polarity of the reference to go from quadrant one to three or four. Taking the polarity of the control voltage and



Note : ZCD — Zero Crossing Detector for polarity detection

Figure 4.22 PWM current-controller implementation with ramp carrier signal

combining it with the polarities of the current and speed gives the operational quadrant. The chopper *on* and *off* pulses generated with the intersection of rectified v_c and carrier ramp will then be combined with the quadrant-selector signals of speed, current, and control-voltage polarities to generate the base-drive signals to the chopper switching devices. An illustration is given in the drive-system simulation section.

Instead of a ramp signal for carrier waveform, a unidirectional sawtooth waveform could be used. It is advantageous in that it has symmetry between the rising and falling sides of the waveform, unlike the ramp signal. Its principle of operation is explained in the following.

The voltage applied to the load is varied within one cycle of the carrier signal. This is illustrated in Figure 4.23. The switching logic is summarized as follows:

$$i_a^* - i_a \geq \text{carrier frequency saw tooth waveform magnitude}, T_p = 1, v_a = V_s \quad (4.61)$$

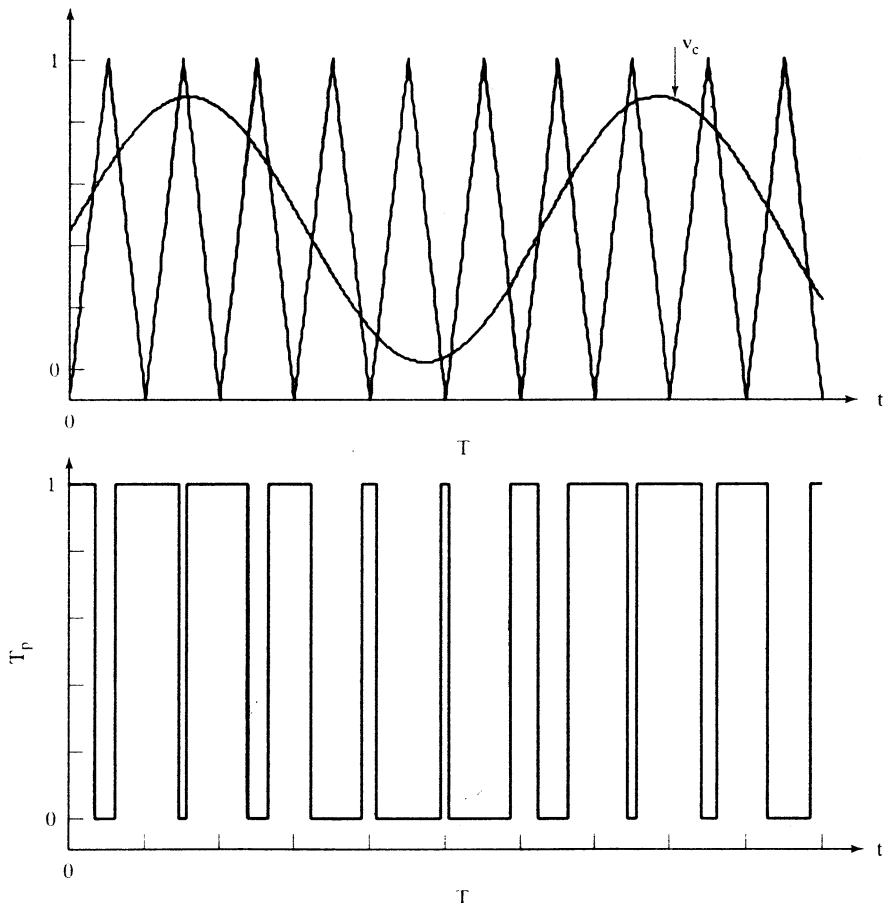


Figure 4.23 Principle of PWM operation

$$i_a^* - i_a < \text{carrier frequency saw tooth waveform magnitude, } T_p = 0, v_a = 0 \quad (4.62)$$

For a fast response, the current error is amplified so that a small current error would activate the chopper control. The PWM controller has the advantage of smaller output ripple current for a given switching frequency, compared to the hysteresis current controller described later.

The pulses generated from the PWM controller are substituted for T_p in the block diagram shown in Figure 4.22. They are then processed for quadrant selection, interlock, and safety features, and appropriate base-drive signals are generated for application to the chopper circuit.

4.12.4 Hysteresis-Current Controller

The PWM current controller acts once a cycle, controlling the duty cycle of the chopper. The chopper then is a variable voltage source with average current control. Instantaneous current control is not exercised in the PWM current controller. In between two consecutive switchings, the current can exceed the maximum limit; if the PWM controller is sampled and held once a switching cycle, then the current is controlled on an average but not on an instantaneous basis. The hysteresis controller overcomes such a drawback by converting a voltage source into a fast-acting current source. The current is controlled within a narrow band of excursion from its desired value in the hysteresis controller. The hysteresis window determines the allowable or preset deviation of current, Δi . Commanded current and actual current are shown in Figure 4.24 with the hysteresis windows. The voltage applied to the load is determined by the following logic:

$$i_a \leq i_a^* - \Delta i, \quad \text{set } v_a = V_s \quad (4.63)$$

$$i_a \geq i_a^* + \Delta i, \quad \text{reset } v_a = 0 \quad (4.64)$$

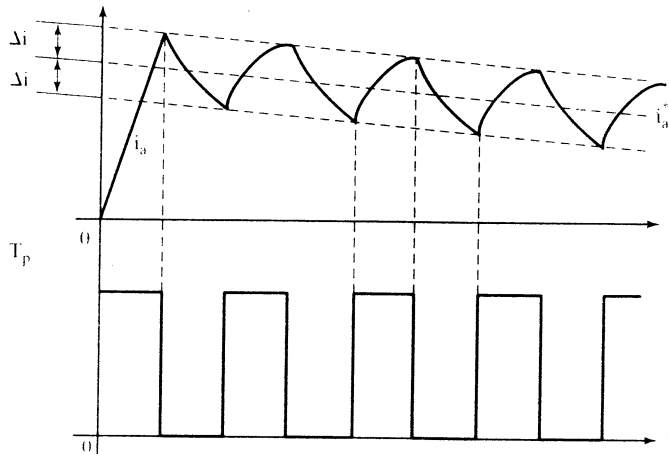


Figure 4.24 Hysteresis-controller operation

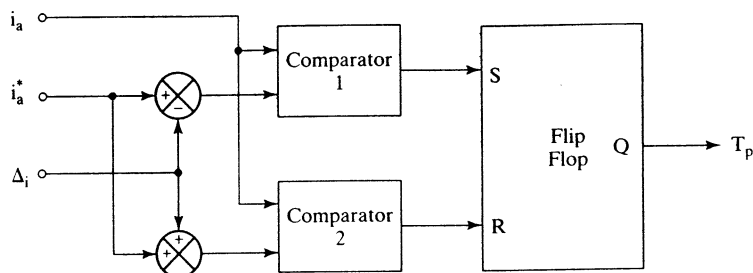


Figure 4.25 Realization of hysteresis controller

TABLE 4.1 Comparison of current controllers

Characteristics	Current Controllers	
	Hysteresis	PWM
Switching Frequency	Varying	Fixed at Carrier Frequency
Speed of Response	Fastest	Fast
Ripple Current	Adjustable	Fixed
Filter Size	Dependent on Δi	Usually small
Switching Losses	Usually high	Low

The realization of this logic is shown in Figure 4.25. The window, Δi , can either be externally set as a constant or be made a fraction of armature current, by proper programming. The chopping frequency is a varying quantity, unlike the constant frequency in the PWM controller. This has the disadvantage of higher switching losses in the devices with increased switching frequency.

The T_p pulses issued from the hysteresis controller replace the block consisting of the comparator output in the Figure 4.22. All other features remain the same for the implementation of the hysteresis controller. This controller provides the fastest response, by means of its instantaneous action. A qualitative comparison of the PWM and hysteresis controllers is summarized in Table 4.1.

4.12.5 Modeling of Current Controllers

The current-error amplifier is modeled as a gain and is given by

$$G_c(s) = K_c \quad (4.65)$$

The chopper is modeled as a first-order lag, with a gain given by

$$G_r(s) = \frac{K_r}{\left(1 + \frac{sT}{2}\right)} \quad (4.66)$$

The PWM current controller has a delay of half the time period of the carrier waveform, and its gain is that of the chopper. Hence, its transfer function, including that of the chopper, is

$$G_c G_r(s) = \frac{K_c K_r}{\left(1 + \frac{sT}{2}\right)} \quad (4.67)$$

where K_c is the gain of the PWM current controller, K_r is the gain of the chopper, and the time constant T is given by,

$$T = \frac{1}{\text{carrier frequency}} = \frac{1}{f_c} \quad (4.68)$$

The gain of the PWM current controller is dependent on the gain of the current-error amplifier. For all practical purposes, the PWM current control loop can be modeled as a unity-gain block if the delay due to the carrier frequency is negligible.

The hysteresis controller has instantaneous response; hence, the current loop is approximated as a simple gain of unity.

4.12.6 Design of Current Controller

The current loop is not easily approximated into a first-order transfer function, unlike the case of the phase-controlled-rectifier drive system. The chopping frequency is considered to be high enough that the time constant of the converter is very much smaller than the electrical time constants of the dc motor. That leads to the converter model given by the product of the converter and current-controller gains. Then the closed-loop current-transfer function is written as

$$\frac{i_a(s)}{i_a^*(s)} = K_c K_r K_1 \frac{(1 + sT_m)}{(1 + sT_1)(1 + sT_2) + H_c K_r K_c K_1 (1 + sT_m)} \quad (4.69)$$

where $K_1 = \frac{B_t}{K_b^2 + R_a B_t}$ and K_c and K_r are the current-controller and chopper gains, respectively. The chopper gain is derived as

$$K_r = \frac{V_s}{V_{cm}} \quad (4.70)$$

where V_s is the dc link voltage and V_{cm} is the maximum control voltage.

The gain of the current controller is not chosen on the basis of the damping ratio, because the poles are most likely to be real ones. Lower the gain of the current controller; the poles will be far removed from the zero. The higher the value of the gain, the closer will one pole move to the zero, leading to the approximate cancellation of the zero. The other pole will be far away from the origin and will contribute to the fast response of the current loop. Consider the worked-out Example 3.4 from the previous chapter. The values of various constants are :

$$\begin{aligned} K_1 &= 0.049 & H_c &= 0.355 & T_m &= 0.7 & T_1 &= 0.1077 \\ T_2 &= 0.0208 & V_s &= 285 \text{ V} & V_{cm} &= 10 \text{ V} & K_r &= 28.5 \text{ V} \end{aligned}$$

The zero of the closed current-loop transfer function is at -1.42 . Note that this is not affected by the current controller. The closed-loop poles for current-controller gains of 0.1, 1, and 10 are given below, along with their steady-state gains, in the following table.

K_c	Poles	Steady-state gain
0.1	$-7.18, -65.11$	21.33
1.0	$-3.24, -203.45$	213.38
10.0	$-1.66, -1549.0$	2580.30

It is seen that, as the gain increases, one of the poles moves closer to the zero at $-\frac{1}{T_m}$, enabling cancellation and a better dynamic response.

In high-performance motor-drive systems, it is usual to have a PI current controller instead of the simple proportional controller illustrated in this section. The PI controller provides zero steady-state current error, whereas the proportional controller will have a steady-state error. In the case of the PI current controller, the design procedure for the phase-controlled dc motor-drive system can be applied here without any changes. The interested reader can refer to Chapter 3 for further details on the design of the PI current controller.

4.12.7 Design of Speed Controller by the Symmetric-Optimum Method

The block diagram for the speed-controlled drive system with the substitution of the current-loop transfer function is shown in Figure 4.26. Assuming that the time constant of the speed filter is negligible, the speed-loop transfer function is derived from Figure 4.26 as

$$\frac{\omega_m(s)}{\omega_r^*(s)} = \frac{1}{H_w} \cdot \frac{a_0(1 + sT_s)}{a_0 + a_1s + a_2s^2 + a_3s^3} \quad (4.71)$$

where

$$a_0 = K_s \frac{K_s}{T_s} \quad (4.72)$$

$$a_1 = 1 + H_c K_r K_c K_l + K_s K_s \quad (4.73)$$

$$a_2 = T_1 + T_2 + H_c K_r K_c K_l T_m \quad (4.74)$$

$$a_3 = T_1 T_2 \quad (4.75)$$

$$K_s = K_b \frac{H_w K_r K_c K_l}{B_t} \quad (4.76)$$

This is very similar to the equation derived in Chapter 3, from which the following symmetric optimum conditions are imposed:

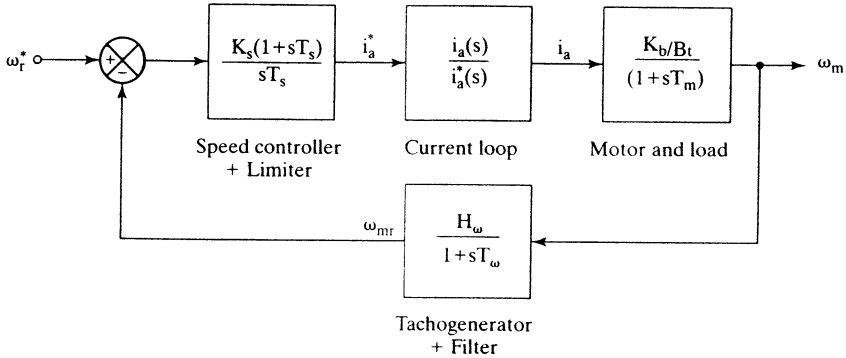


Figure 4.26 Simplified speed-controlled dc motor drive fed from a chopper with hysteresis-current control

$$a_1^2 = 2a_0a_2 \quad (4.77)$$

$$a_2^2 = 2a_1a_3 \quad (4.78)$$

Conditions given by equations (4.77) and (4.78) result in speed-controller time and gain constants given by

$$K_s = \frac{1}{K_5} \cdot \left\{ \frac{a_2^2}{2a_3} - (1 + H_c K_r K_c K_1) \right\} \quad (4.79)$$

$$T_s = \frac{2K_5 K_s a_2}{a_1^2} \quad (4.80)$$

For the same example, the speed-controller gain and time constants for various gains of the current controller are given next, together with the closed-loop poles and zeros and the steady-state gains of the speed-loop transfer function.

Current controller gain	Speed controller gain	Speed controller time constant	Steady-state gain	Zero	Poles
0.1	106.7	0.045	1628	-22.22	-18.3 ± j31, -36.9
1.0	102.9	0.0188	38,000	-53.2	-52.7 ± j88.9, -105.1
10.0	597.0	0.0026	1.6 × 10 ⁷	-384.6	-393 ± j666.8, -792.4

Increasing current-controller gain has drastically reduced the speed-loop time constant without appreciably affecting the damping ratio of the closed-loop speed-control system. Because of the large integral gain in the speed controller, its output will saturate in time. An anti-windup circuit is necessary to overcome the saturation in this controller design and thus to keep the speed controller responsive. The anti-windup circuit can be realized in many ways. One of the implementations is shown in Figure 4.27.

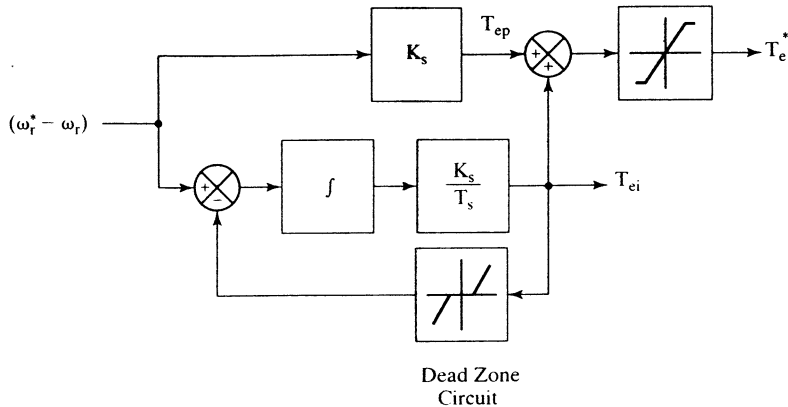


Figure 4.27 Anti-windup circuit with PI speed controller

The saturation due to the integral action alone is countered in this implementation. It is achieved by a negative-feedback control of the integral-controller output through a dead-zone circuit. This dead-zone circuit produces an output only when the integral controller output exceeds a preset absolute maximum, i.e., when the controller output saturates. This feedback is subtracted from the speed error and the resulting signal constitutes the input to the integrator. When the integral-controller output saturates, the input to the integral controller is reduced. This action results in the reduction of the integral controller's output, thus pulling the integral controller from saturation and making the controller very responsive. If there is no saturation of the integral-controller output, then the feedback is zero (because of the dead-zone circuit); hence, the anti-windup circuit is inactive in this implementation. The outputs of the integral and the proportional controllers are summed, then limited, to generate the torque reference signal. By keeping the outputs of the proportional and integral controllers separate, their individual tuning and the beneficial effect of the high proportional gain are maintained.

4.13 DYNAMIC SIMULATION OF THE SPEED-CONTROLLED DC MOTOR DRIVE

A dynamic simulation is recommended before a prototype or an actual drive system is built and integrated, to verify its capability to meet the specifications. Such a simulation needs to consider all the motor-drive elements, with nonlinearities. The transfer-function approach could become invalid, because of the nonlinear current loop. This drawback is overcome with the time-domain model developed below. The speed-controlled drive shown in Figure 4.20, but modified to include the effects of field excitation variation and with a PWM or a hysteresis controller, is considered for the simulation.

4.13.1 Motor Equations

The dc motor equations, including its field, are

$$V_a = R_a i_a + L_a \frac{di_a}{dt} + K\Phi_f \omega_m \quad (4.81)$$

$$V_f = R_f i_f + L_f \frac{di_f}{dt} \quad (4.82)$$

$$T_e - T_l = J \frac{d\omega_m}{dt} + B_l \omega_m \quad (4.83)$$

$$T_e = K\Phi_f i_a \quad (4.84)$$

where

$$K\Phi_f = M i_f \quad (4.85)$$

and M is the mutual inductance between the armature and field windings.

The state variables are defined as

$$x_1 = i_a \quad (4.86)$$

$$x_2 = \omega_m \quad (4.87)$$

$$x_3 = i_f \quad (4.88)$$

The motor equations in terms of the state variables are

$$\dot{x}_1 = -\frac{R_a}{L_a} x_1 - \frac{M}{L_a} x_2 x_3 + \frac{1}{L_a} V_a \quad (4.89)$$

$$\dot{x}_2 = \frac{M}{J} x_1 x_3 - \frac{B_l}{J} x_2 - \frac{T_l}{J} \quad (4.90)$$

$$\dot{x}_3 = -\frac{R_f}{L_f} x_3 + \frac{1}{L_f} V_f \quad (4.91)$$

4.13.2 Speed Feedback

The tachogenerator and the filter are combined in the transfer function as

$$\frac{\omega_{mr}(s)}{\omega_m(s)} = \frac{H_\omega}{1 + sT_w} \quad (4.92)$$

The state diagram of equation (4.92) is shown in Figure 4.28, where

$$x_4 = \omega_{mr} \quad (4.93)$$

and the state equation is obtained from the state diagram as

$$\dot{x}_4 = \frac{1}{T_w} (H_\omega x_2 - x_4) \quad (4.94)$$

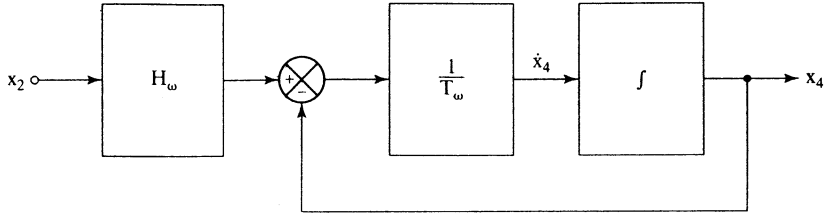


Figure 4.28 State diagram of the speed feedback

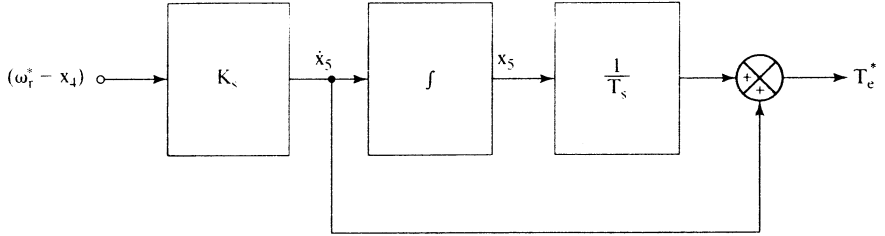


Figure 4.29 State diagram of the speed controller

4.13.3 Speed Controller

The input to the speed controller is the speed error, and the controller is of a PI type given by

$$\frac{T_e^*(s)}{(\omega_r^* - \omega_{mr})} = \frac{K_s(1 + sT_s)}{sT_s} \quad (4.95)$$

The state diagram of the speed controller is shown in Figure 4.29, where

$$y_1 = (\omega_r^* - \omega_{mr}) \quad (4.96)$$

The state equation and the torque equation are

$$\dot{x}_5 = K_s y_1 = K_s(\omega_r^* - \omega_{mr}) = K_s(\omega_r^* - x_4) \quad (4.97)$$

$$T_e^* = \frac{x_5}{T_s} + \dot{x}_5 = \frac{x_5}{T_s} + K_s y_1 = \frac{x_5}{T_s} + K_s(\omega_r^* - x_4) = -K_s x_4 + \frac{1}{T_s} x_5 + K_s \omega_r^* \quad (4.98)$$

There is a limit on the maximum torque command, which is included as a constraint as

$$-T_{e^*}(\max) \leq T_e^* \leq +T_{e^*}(\max) \quad (4.99)$$

4.13.4 Command Current Generator

The current command is calculated from the torque command and is given as

$$i_a^* = \frac{T_e^*}{M i_f} = \frac{T_e^*}{M} \cdot \frac{1}{x_3} \quad (4.100)$$

Substituting for T_c^* from equation (4.98) into (4.100) gives the current command as

$$i_a^* = \left(-\frac{K_s}{M}\right) \cdot \frac{x_4}{x_3} + \left(\frac{1}{T_s M}\right) \cdot \frac{x_5}{x_3} + \frac{K_s}{M} \cdot \frac{\omega_r^*}{x_3} \quad (4.101)$$

4.13.5 Current Controller

The current error is

$$i_{cr} = i_a^* - i_a \quad (4.102)$$

which, in terms of the motor and controller parameters, is

$$i_{cr} = \left(-\frac{K_s}{M}\right) \cdot \frac{x_4}{x_3} + \left(\frac{1}{T_s M}\right) \cdot \frac{x_5}{x_3} + \left(\frac{K_s}{M} \cdot \omega_r^*\right) \cdot \frac{1}{x_3} - x_1 \quad (4.103)$$

The current-controller logic is illustrated for hysteresis control. It is given as

$$i_{cr} > \Delta I, \quad \text{set } V_a = V_s \quad (4.104)$$

$$i_{cr} < -\Delta I, \quad \text{reset } V_a = 0 \quad (4.105)$$

Note that the control logic considers only the forward-motoring mode. Suitable logic has to be incorporated for the rest of the quadrants' operation. For PWM control, the logic given by equations (4.61) and (4.62) is used.

4.13.6 System Simulation

The anti-windup controller for the speed controller can be modelled and can be incorporated in place of the PI speed controller. For the example illustrated in the following pages, the anti-windup circuit is not incorporated with the PI speed controller. Equations (4.89), (4.90), (4.91), (4.94), and (4.97) constitute the state equations; equations (4.100), (4.104), and (4.105) are to incorporate the limits and current controller. The solution of the state equations is achieved by numerical integration, with ω_r^* serving as the only input into the block diagram. A flowchart for the dynamic simulation is given in Figure 4.30.

Typical dynamic performance of a speed-controlled chopper-fed dc motor drive is shown in Figure 4.31. The current controller is of hysteresis type, and a 0.25-p.u. steady-state load torque is applied. 0.5 p.u. step speed is commanded, with a maximum torque limit of 2 p.u. resulting in the armature current limit of 2 p.u. The current window is 0.1 p.u., sufficient to cause low switching frequency for this drive system. The references are shown with dotted lines. The torque response is obtained in 1.3 ms. and the speed response is uniform and without any noticeable oscillations. Note that, as the rotor speed reaches the commanded speed, the speed error (and hence the electromagnetic torque command) decreases. During this time, to reduce the electromagnetic torque, only zero voltage is applied across the armature terminals. When the armature current goes below its reference value, then the source voltage is applied to the armature.

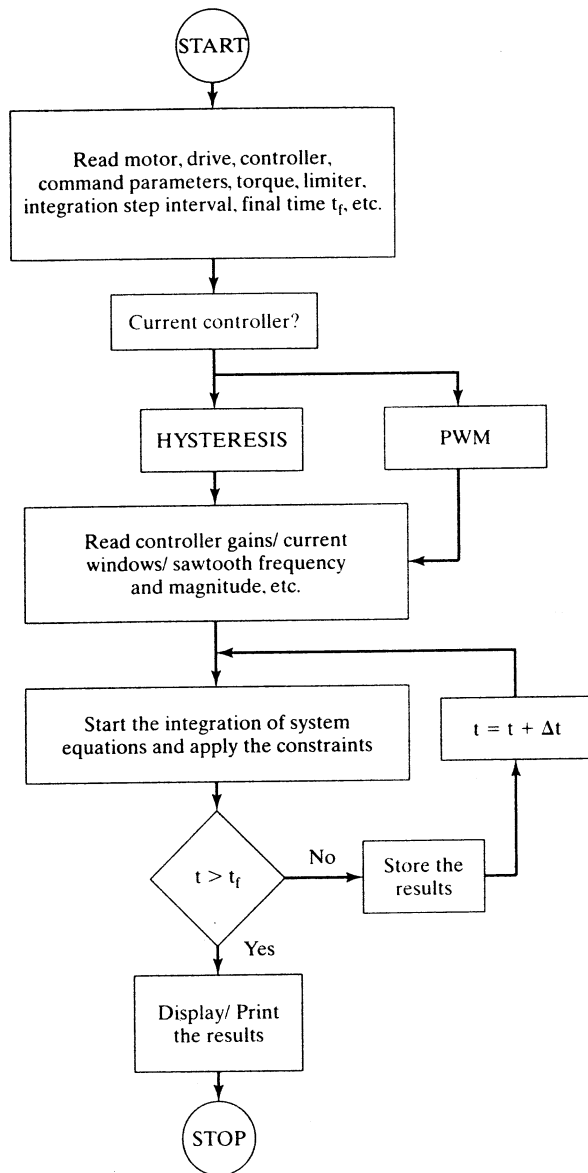


Figure 4.30 Flowchart for the dynamic simulation of the chopper-controlled dc motor drive

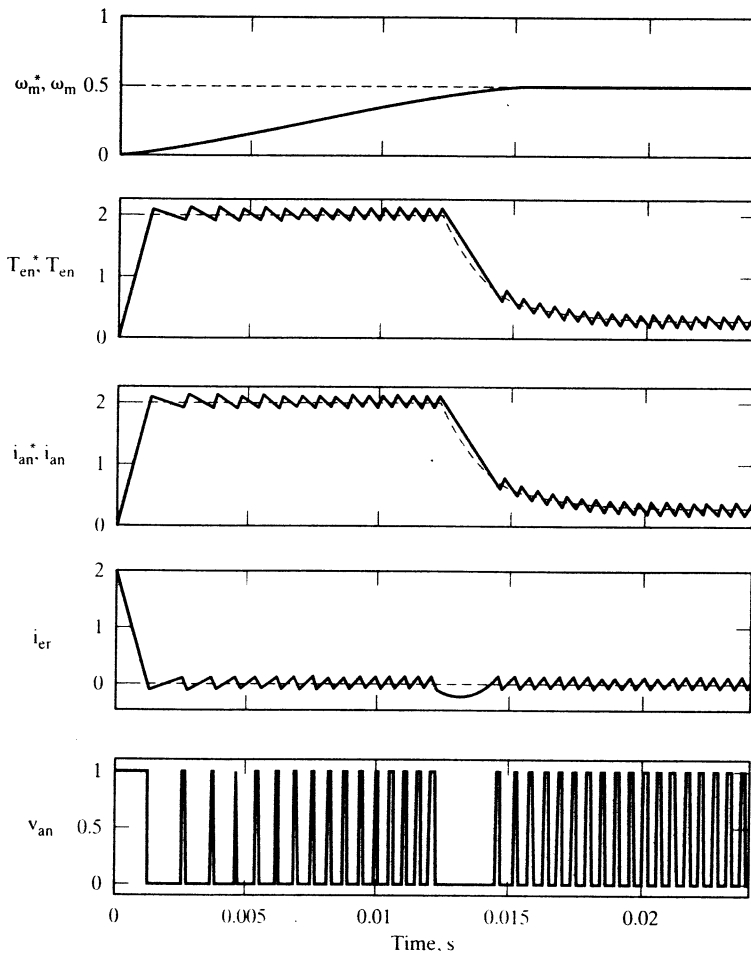


Figure 4.31 Dynamic performance of a one-quadrant chopper-controlled separately-excited dc motor drive for a step command in speed reference, in normalized units

Figure 4.32 shows the performance of a four-quadrant drive system with hysteresis-current controller, for a load torque of 0.25 p.u.. The reference speed is step-commanded, so the speed error reaches a maximum of 2 p.u., as in the first-quadrant operation described in Figure 4.31. When the speed reference goes to zero and then to negative speed, the torque command follows, but the armature current is still positive. To generate a negative torque, the armature current has to be reversed, i.e., in this case to negative value. The only way the current can be reversed is by taking it through zero value. To reduce it to zero, the armature current is made to charge the dc link by opening the transistors, thus enabling the diodes across the other pair of nonconducting transistors to conduct. This corresponds to part of the

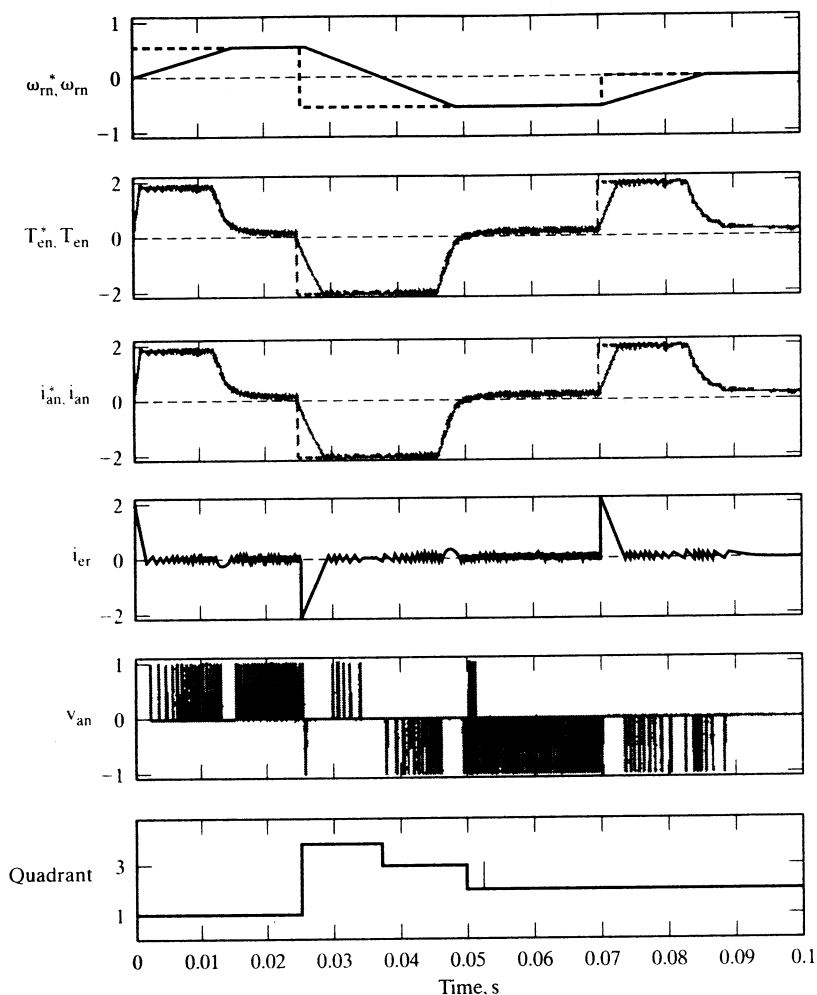


Figure 4.32 Dynamic performance of a four-quadrant chopper-controlled separately-excited dc motor drive for a step command in bipolar speed reference, in normalized units

second-quadrant operation, as shown in Figure 4.7, with the diodes D_3D_4 conducting. When the current is driven to zero, note that the speed, and hence the induced emf, are still positive. The current is reversed with the aid of the induced emf by switching T_4 only, as in to the fourth-quadrant operation shown in Figure 4.10. The induced emf enables a negative current through D_2 , the machine armature, and T_4 . The current rises fast; when it exceeds the current command by the hysteresis window, switch T_4 is turned off, enabling D_1 to conduct along with D_2 . The armature current charges the dc source, and the voltage applied across the machine is $+V_s$ during

this instant. If the current falls below the command by its hysteresis window, then the armature is shorted with T_4 again, to let the current build up. This priming and charging cycle continues until the speed reaches zero. The drive algorithm for the part discussed is briefly summarized as follows.

ω_r^*	T_c^*	ω_r	i_a^*	i_a	Condition
+	+	+	+	+	If $\omega_r < \omega_r^*$, $i_{cr} \geq \Delta i$, set $V_a = V_s$ $i_{cr} \leq -\Delta i$, reset $V_a = 0$ (Quadrant I)
+	-	+	-	+	If $\omega_r > \omega_r^*$, $V_a = -V_s$ until $i_a \rightarrow 0$ (Quadrant I)
+	-	+	-	-	If $\omega_r > \omega_r^*$, $i_{cr} \geq \Delta i$, set $V_a = V_s$ $i_{cr} \leq -\Delta i$, reset $V_a = 0$ (Quadrant IV)

To reverse the rotor speed, the negative torque generation has to be continued. When the stored energy has been depleted in the machine as it reaches stand-still, energy has to be applied to the machine, with the armature receiving a negative current to generate the negative torque. This is achieved by the switching of T_3 and T_4 . From now on, it corresponds to third- and second-quadrant operations for the machine to operate in the reverse direction at the desired speed and to bring it to zero speed, respectively. For the operation in the reverse direction, the input armature voltage conditions can be derived as in the table given above. The quadrants of operation are also plotted in the figure, to appreciate the correlation of speed, torque, voltage, and currents in the motor drive system.

4.14 APPLICATION

Forklift trucks are used in material-handling applications, such as for loading, unloading, and transportation of materials and finished goods in warehouses and manufacturing plants. The forklift trucks are operated by rechargeable battery-driven dc series motor-drive systems. Such electrically operated equipment is desired, to comply with zero pollution in closed workplaces and to minimize fire hazards. The dc series motor for a typical forklift [6] is rated at 10 hp, 32 Vdc, 230 A, 3600 rpm and is totally enclosed and fan-cooled. It operates with a duty cycle of 20%. Four-quadrant operation is desired, to regenerate and charge the batteries during braking, and both directions of rotation are required, for moving in the forward and reverse directions. The torque is increased by having a gear box between the motor and the forklift drive train. A schematic diagram of the forklift control is shown in Figure 4.33. The presence of an operator means that the forklift has no automatic closed-loop speed control; the operator provides the speed feedback and control by continuous variation of speed reference.

Chopper-controlled dc motor drives are also used in conveyors, hoists, elevators, golf carts, people carriers in airport lobbies, and some variable-speed hand tools.

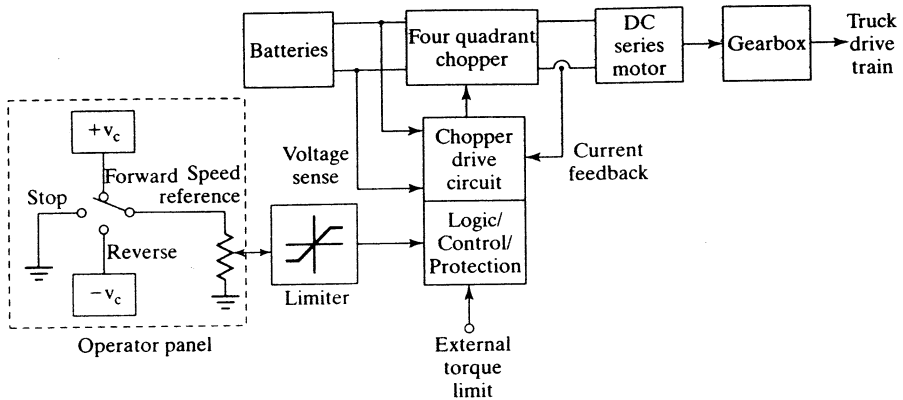


Figure 4.33 Block-diagram control schematic of a battery-fed chopper-controlled dc series motor-drive system

Example 4.6

A separately-excited chopper-controlled dc motor drive is considered for a paper winder in a winding-unwinding controller. This application uses essentially constant horsepower because the tension of the strip decreases as the build-up increases on the roll; hence, the winder can be speeded up. The maximum increase in speed is determined by the output power of the drive system. The motor ratings and parameters are as follows: 200 kW, 600 V, 2400 rpm, $R_a = 0.05 \Omega$, $L_a = 0.005 \text{ H}$, $K_b = 2.32 \text{ V/rad/sec}$, $B_l = 0.05 \text{ N-m/rad/sec}$, $J = 100 \text{ K g-m}^2$, $R_f = 30 \Omega$, $L_f = 20 \text{ H}$, Input = 460 V $\pm 10\%$, 3 ph, 60 Hz $\pm 3 \text{ Hz}$.

Design the chopper power circuit with a current capacity of 2 p.u. for short duration and dc link voltage ripple factor of 1% for dominant harmonic. (The ripple factor is the ratio between the rms ac ripple voltage and the average dc voltage.)

Solution

$$\omega_b = \frac{2\pi \times 2400}{60} = 251.33 \text{ rad/sec}$$

$$T_b = \frac{P_b}{\omega_b} = \frac{200 \times 1000}{251.33} = 795.77 \text{ N-m}$$

$$I_b = \frac{T_b}{K_b} = \frac{795.77}{2.32} = 343.01 \text{ A}$$

$$I_{\max} = 2 \times I_b = 686.02 \text{ A}$$

Consider the power circuit shown in Figure 4.11.

(i) DC Diode-Bridge Rectifier

Considering the maximum ac input voltage,

$$\text{Minimum peak inverse voltage} = \sqrt{2}(V + 0.1V) = \sqrt{2} \times 460 \times 1.1 = 715.6 \text{ V}$$

$$\text{RMS current per diode (approximate)} = I_{\max} \sqrt{\frac{1}{3}} = (680) \sqrt{\frac{1}{3}} = 392.6 \text{ A}$$

(ii) DC Link Filter

The dominant input harmonic to the dc link voltage is the sixth. Its rms value is

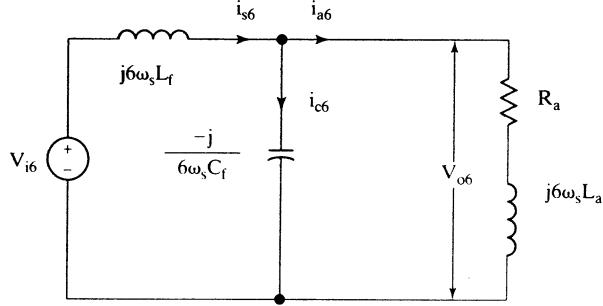


Figure 4.34 Harmonic-equivalent circuit of the dc motor drive

$$V_{i6} = \frac{1}{\sqrt{2}} \left(\frac{3\sqrt{2}V}{\pi} \cdot \frac{2}{35} \right)$$

from equation (3.175), which when substituted for nominal line-to-line voltage of 460 V, gives

$$V_{i6} = 25.1 \text{ V}$$

$$\text{Given ripple factor} = 0.01 = \frac{V_{o6}}{V_{dc}}$$

where V_{o6} is the output voltage ripple across the capacitor and V_{dc} is the average dc voltage.

$$V_{o6} = 0.01V_{dc} = 0.01 \times 1.35 \times V = 0.01 \times 1.35 \times 460 = 6.21 \text{ V}$$

Hence, the ratio of sixth-harmonic output voltage to input voltage is given as

$$\frac{V_{o6}}{V_{i6}} = \frac{6.21}{25.1} = 0.247$$

The sixth-harmonic equivalent circuit for the filter and motor load is shown in Figure 4.34.

From the figure, it is seen that, to minimize the harmonic current to the load, the capacitive reactance has to be approximately 10 times smaller than the load impedance:

$$\sqrt{R_a^2 + (6\omega_s L_a)^2} \gg \frac{1}{6\omega_s C_f} \cong \frac{10}{6\omega_s C_f}$$

from which C_f is evaluated as

$$C_f = \frac{10}{6\omega_s \sqrt{R_a^2 + (6\omega_s L_a)^2}} = \frac{10}{(6 \times 358.14) \sqrt{(0.05)^2 + (6 \times 358.14 \times 0.005)^2}} = 433 \text{ } \mu\text{F}$$

where ω_s corresponds to the lowest line frequency, i.e.,

$$2\pi(f_s - 3) = 2\pi(60 - 3) = 358.14 \text{ rad/sec}$$

The load impedance is 10 times the capacitive reactance; for the purpose of finding the ratio between the capacitive voltage and the input source voltage, then, the load impedance can be treated as an open circuit. This gives rise to

$$\frac{V_{o6}}{V_{i6}} = \frac{\frac{1}{n\omega_s C_f}}{n\omega_s L_f - \frac{1}{n\omega_s C_f}} = \frac{1}{n^2\omega_s^2 L_f C_f - 1} = 0.247$$

from which $L_f C_f$ is obtained as

$$L_f C_f = \frac{\frac{1}{0.247} + 1}{n^2 \omega_s^2} = 1.09 \times 10^{-6}$$

Hence, the filter inductance is obtained, by using the previously calculated C_f , as

$$L_f = \frac{1.09 \times 10^{-6}}{C_f} = \frac{1.09 \times 10^{-6}}{433 \times 10^{-6}} = 2.52 \text{ mH}$$

(iii) Chopper Device Ratings

$$I_T = I_{\max} = 686 \text{ A}$$

$$V_D = V_T = \sqrt{2}(V + 0.1V) = 715.6 \text{ V}$$

(iv) Device Selection

IGBT.

A single device per switch is sufficient in both these categories.

4.15 SUGGESTED READINGS

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4.16 DISCUSSION QUESTIONS

1. Why is the duty cycle usually changed by varying the on-time rather than the chopping frequency?
2. Very small duty cycles near zero are not possible to realize in practice. What are the constraints in such cases?
3. Similarly, a duty cycle of one is not feasible in the chopper. What is the reason?
4. Draw the gating pulses for each of the four quadrants of operation of the chopper-controlled dc motor drive.
5. Is a sine-wave output current possible from a chopper circuit?
6. Is a sine-wave voltage output possible from a chopper circuit?
7. The chopper can be switched at very high frequencies. What are the factors that limit high-frequency operation?
8. A 5-hp dc motor is to be driven from a chopper for two applications: a robot, and a winder. Discuss the power device to be selected for each of the applications.
9. Compare a four-quadrant chopper and a phase-controlled converter fed from a 3-phase ac supply with regard to the number of power devices, speed of response, control complexity, and harmonics generated.
10. A two-quadrant drive to operate in FM and FR is required. Which one has to be recommended between Figures 4.13(ii) and (iii)?
11. The averaging and the instantaneous steady-state computation techniques give identical results in the continuous-conduction mode. Is this true for the discontinuous current mode, too?
12. Ripple current is minimized by either increasing the chopper frequency or including an inductance in series with the armature of the dc motor. Discuss the merits and demerits of each alternative. Are the alternatives constrained by the size of the motor drive?
13. Discuss the control circuit design for a two-quadrant chopper circuit.
14. Discuss the impact of the choice of the current controllers on the dynamic performance of the dc motor-drive system.
15. Design a digital controller for a hysteresis-current controller.
16. Design an analog version of the PWM current controller. Use commercially available integrated chips and operational amplifiers in the design.
17. Which one of the current controllers produces the highest switching losses and motor copper losses?
18. Is an innermost voltage loop necessary for fast response of the dc motor drive?
19. The hysteresis window, Δi , can be made a constant or a variable. Discuss the merits and demerits of both alternatives.
20. Likewise, the PWM carrier frequency can be either a constant or a variable. Discuss the merits and demerits of both alternatives.
21. The window in the hysteresis controller and the carrier frequency in the PWM controller are made to be variable quantities. What are they to be dependent upon for variation?
22. Is the parameter sensitivity of the open-loop chopper-controlled dc motor drive different from that of the phase-controlled dc motor drive?
23. Time-domain simulation can eliminate the errors in the final construction and testing of a chopper-controlled dc motor drive. Discuss a method of including the front-end uncontrolled rectifier and filter in the simulation.

24. The time-domain dynamic model derived and developed in this chapter has to be very slightly modified to treat either a series or a cumulatively-excited dc motor drive. What are the changes to be introduced ?
25. The transfer-function model is true for both small- and large-signal analysis of the separately-excited dc motor drive. A chopper-controlled dc series motor drive has to be considered for its speed-controller design. Will the same block diagram be adequate for the task at hand?

4.17 EXERCISE PROBLEMS

1. A chopper is driving a separately-excited dc motor whose details are given in Example 4.1. The load torque is frictional and delivers rated torque at rated speed. Calculate and plot the duty cycle vs. speed. The maximum electromagnetic torque allowed in the motor is 2 p.u. The friction coefficient of the motor is $0.002 \text{ N}\cdot\text{m}/\text{rad}/\text{sec}$.
2. Consider the dc motor chopper details given in Example 4.2. Compute the torque-speed characteristics for a duty cycle of 0.4, using the averaging and instantaneous steady state computation techniques. Do the methods compare well in the discontinuous current-conduction mode?
3. Compare the switch ratings of the two-quadrant choppers shown in Figures 4.13(ii) and (iii).
4. A chopper-controlled dc series motor drive is intended for traction application. Calculate its torque-speed characteristics for various duty cycles. The motor details are as follows: 100 hp, 500 V, 1500 rpm, $R_a + R_f = 0.01 \Omega$, $L_a + L_{se} = 0.012 \text{ H}$, $M = 0.1 \text{ H}$, $J = 3 \text{ Kg} \cdot \text{m}^2$, $B_f = 0.1 \text{ N}\cdot\text{m}/\text{rad}/\text{sec}$. The chopper has an input source voltage of 650 V and operates at 600 Hz.
5. The motor chopper given in Example 4.4 is used in a position servo application. It is required that the pulsating torque be less than 1% of the rated torque. At zero speed, the motor drive is producing a maximum of 3 p.u. torque. Determine (i) the increase in switching frequency and (ii) the value of series inductance to keep the pulsating torque within the specification.
6. The above problem has an outermost position loop. The position controller is of PI type. Determine the overall-position transfer function and find its bandwidth for $K_p = 1$ and $T_p = 0.2 \text{ sec}$, if the system is stable. Assume that the speed controller is of proportional type, with a gain of 100. The current controller is a hysteresis controller.
7. A chopper-controlled dc motor has the following parameters:
 6.3 A , 200 V , 1000 rpm , $R_a = 4 \Omega$, $L_a = 0.018 \text{ H}$, $K_b = 1.86 \text{ V}/\text{rad}/\text{sec}$, $J = 0.1 \text{ kg} \cdot \text{m}^2$,
 $B_f = 0.0162 \text{ N}\cdot\text{m}/\text{rad}/\text{sec}$, $f_c = 500 \text{ Hz}$, $V_s = 285 \text{ V}$

Determine the following:

- (i) Torque-speed characteristics for duty cycles of 0.2, 0.4, 0.6, and 0.8 in the forward-motoring mode;
- (ii) The average currents at 500 rpm, using averaging and instantaneous steady-state evaluation techniques; (Assume a suitable duty cycle.)
- (iii) Critical duty cycle vs. speed, with and without an external inductance of 20 mH. (Draw it.)

8. Determine the speed-controller gain and time constant for the following separately-excited chopper-controlled dc motor drive:

250 hp, 500 V, 1250 rpm, 92% Efficiency, $R_a = 0.052 \, \Omega$, $L_a = 1 \, \text{mH}$,
 $K_b = 3.65 \, \text{V/rad/sec}$, $J = 5 \, \text{kg}\cdot\text{m}^2$, $B_l = 0.2 \, \text{N}\cdot\text{m/rad/sec}$, $V_s = 648 \, \text{V}$.

The current controller has a PWM strategy with a carrier frequency of 2 kHz. Calculate also the speed response for a speed command of 0 to 0.8 p.u., when the load torque is maintained at 0.5 p.u. {Hint: Calculate the speed controller gain and time constants, assuming $B_l = 0$.}

Polyphase Induction Machines

5.1 INTRODUCTION

A brief introduction to the theory and principle of operation of induction machines is given in this chapter. The notations are introduced, and consistent use of them in the text is adhered to. The principle of operation of the induction motor is developed to derive the steady-state equivalent circuit and performance calculations. They are required to evaluate the steady-state response of variable-speed induction-motor drives in the succeeding chapters. The dynamic simulation is one of the key steps in the validation of the design process of the motor-drive systems, eliminating inadvertent design mistakes and the resulting errors in the prototype construction and testing and hence the need for dynamic models of the induction machine. The dynamic model of the induction machine in direct-, quadrature-, and zero-sequence axes (known as dqo axes) is derived from fundamentals. This is generally considered difficult and is avoided by many practicing engineers. Hence, care is taken to maintain simplicity in the derivation, while physical insight is introduced. The dqo model uses two windings for each of the stator and rotor of the induction machine. The transformations used in the derivation of various dynamic models are based on simple trigonometric relationships obtained as projections on a set of axes. The dynamic model is derived with the frames of observation rotating at an arbitrary speed. The most useful models in stationary, rotor, and synchronous reference frames are obtained as particular cases of the arbitrary reference-frames model. The dynamic model is used to obtain transient responses, small-signal equations, and a multitude of transfer functions, all of which are useful in the study of converter-fed induction-motor drives. Space-phasor approach has further simplified the polyphase induction machine model to one equivalent stator and one rotor winding, thereby evoking a powerful similarity to the dc machine to correspond with its armature and field windings. The space-phasor modeling of the induction motor is introduced as a simple extension of dqo models. The merits and demerits of the space-phasor represen-

Note that the slip does not enter in this; the ratio between the stator and rotor induced emfs viewed at stator frequency does not contain it, and rotor impedance absorbs this slip. Substitution of equation (5.14) into equation (5.13) yields

$$I_r = \frac{E_1}{\frac{(a^2 R_{rr})}{s} + j\omega_s(a^2 L_{rr})} = \frac{E_1}{\frac{R_r}{s} + j\omega_s L_{lr}} \quad (5.16)$$

where R_r and L_{lr} are the stator-referred rotor resistance and leakage inductance, respectively, given as

$$\begin{aligned} R_r &= a^2 R_{rr} \\ L_{lr} &= a^2 L_{lrr} \end{aligned} \quad (5.17)$$

As the fictitious rotor and stator at the air gap have the same induced emf, E_1 , the physical isolation can be removed to get a connected circuit. The final equivalent circuit, referred to the stator, is shown in Figure 5.3. Magnetization is accounted for by the magnetizing branch of the equivalent circuit, consisting of the magnetizing inductance that is lossless and hence cannot represent core losses. An equivalent resistance can represent the core losses. This core-loss resistance is in parallel to the magnetizing inductance, because the core losses are dependent on the flux and hence proportional to the flux linkages and the resulting air gap voltage E_1 . The self-inductances of the stator phase winding, L_s , and of the rotor phase winding referred to the stator, L_r , are obtained as the sum of the magnetizing inductance and respective leakage inductances:

$$L_s = L_m + L_{ls} \quad (5.18)$$

$$L_r = L_m + L_{lr} \quad (5.19)$$

The corresponding reactances are obtained by multiplying the inductances by the stator angular frequency. A simple phasor diagram of the induction motor is drawn by using the equivalent circuit given in Figure 5.3. The basis is provided by the following relationships, contained in equations from (5.20) to (5.25).

When the machine is energized with no load on the rotor, then the rotor circuit is open-circuited, because the slip is zero. The stator current drawn during this condition, known as the no-load current, contributes to the magnetization of the machine

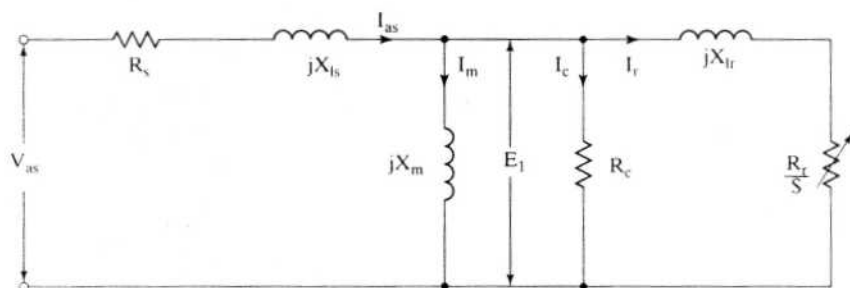


Figure 5.3 Equivalent circuit with the rotor at stator frequency

and resulting core losses. Then this no-load current is viewed as the sum of the magnetizing and core-loss components of the current and accordingly is written as

$$I_o = I_m + I_c \quad (5.20)$$

where I_o is the no-load phase current, I_m is the magnetizing current, and I_c is the core-loss current. The magnetizing current in terms of the air gap voltage and the magnetizing reactance is written as

$$I_m = \frac{E_1}{jX_m} \quad (5.21)$$

where E_1 is the air gap voltage and X_m is the magnetizing reactance. Similarly, the core-loss component of the stator current is written as

$$I_c = \frac{E_1}{R_c} \quad (5.22)$$

where R_c is the resistance to account for the core losses. The rotor phase current is given by

$$I_r = \frac{E_1}{\frac{R_r}{s} + jX_{lr}} \quad (5.23)$$

where I_r is the rotor phase current. The stator phase current then is

$$I_{as} = I_r + I_o \quad (5.24)$$

In terms of the induced emf, stator current, and stator parameters, the applied stator voltage is expressed as the sum of the induced emf and stator impedance voltage drop and is given by

$$V_{as} = E_1 + (R_s + jX_{ls})I_{as} \quad (5.25)$$

where V_{as} is the stator phase voltage.

The phasor diagram is shown in Figure 5.4. The angle ϕ between the stator input voltage and the stator current is the power factor angle of the stator, and λ_m is the mutual flux. The steady-state performance is taken up next by using the equivalent circuit.

5.4 STEADY-STATE PERFORMANCE EQUATIONS OF THE INDUCTION MOTOR

The key variables in the machine are the air gap power, mechanical and shaft output power, and electromagnetic torque. These are derived from the equivalent circuit of the induction machine as follows. The real power transmitted from the stator, P_i , to the air gap, P_a , is the difference between total input power to the stator windings and copper losses in the stator and is given as

$$P_a = P_i - 3I_s^2 R_s \quad (W) \quad (5.26)$$

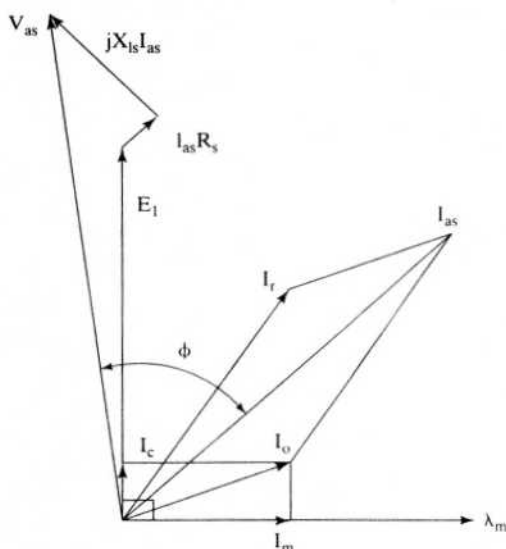


Figure 5.4 Simplified phasor diagram of the induction motor

Neglecting the core losses, the air gap power is equal to the total power dissipated in R_r/s in the three phases of the machine; there is no other element to consume power in the rotor equivalent circuit. It is given as

$$P_a = 3I_r^2 \frac{R_r}{s} \quad (5.27)$$

which could be written alternatively as

$$P_a = 3I_r^2 R_r + 3I_r^2 R_r \frac{(1-s)}{s} \quad (5.28)$$

The $I_r^2 R_r$ term is recognized as the rotor copper loss, and hence the remainder, $I_r^2 R_r \frac{(1-s)}{s}$, gives the power converted into mechanical form. The rotor copper losses are equal to the product of the slip and air gap power from equation (5.27), and this is referred to as slip power. The common term of three in equations (5.27) and (5.28) accounts for the number of phases in the machines, which throughout this text is taken as three. The mechanical power output, P_m , is obtained as

$$P_m = 3I_r^2 R_r \frac{(1-s)}{s} \quad (\text{W}) \quad (5.29)$$

Alternately, in terms of the electromagnetic torque and rotor speed, the mechanical power output is equal to their product:

$$P_m = T_e \omega_m \quad (5.30)$$

where T_e is the internal or electromagnetic torque, derived from equations (5.29) and (5.30) as

$$T_e = \frac{3I_r^2 R_r (1 - s)}{s \omega_m} \quad (5.31)$$

Substituting for the rotor speed in terms of the slip and stator frequency, given by

$$\omega_m = \frac{\omega_r}{P/2} = \frac{\omega_s (1 - s)}{P/2} \quad (5.32)$$

into (5.29), the electromagnetic or air gap torque is obtained as

$$T_e = 3 \left(\frac{P}{2} \right) \frac{I_r^2 R_r}{s \omega_s} \quad (5.33)$$

To obtain the shaft output power of the machine, P_s , the windage and friction losses of the rotor, denoted as P_{fw} , have to be subtracted from the mechanical output power of the machine, symbolically given as follows:

$$P_s = P_m - P_{fw} \quad (5.34)$$

The friction and windage losses are two distinct and separate losses; they are proportional to the speed and the square of the speed, respectively, therefore they have to be represented as a function of speed for evaluation of the variable-speed performance of the induction motor. There are also losses due to stray magnetic fields in the machine; they are covered by the term *stray load* losses. The stray-load losses vary from 0.25 to 0.5 percent of the rated machine output. The stray-load losses are obtained from the measurements on the machine under load from the remainder of the difference between the input power and the sum of the known losses such as the stator and rotor copper and core losses, friction and windage losses, and power output. Note that the stray-load losses have not been accounted for in the equivalent circuit of the machine. Various analytical formulae and empirical relationships are in use, but a precise prediction of the stray losses is very difficult.

Example 5.1

- (i) Find the efficiency of an induction motor operating at full load. The machine details are given in the following:

2000 hp, 2300 V, 3 phase, Star connected, 4 pole, 60 Hz, Full load slip = 0.03746

$R_s = 0.02 \Omega$; $R_r = 0.12 \Omega$; $R_c = 451.2 \Omega$; $X_m = 50 \Omega$; $X_{ls} = X_{lr} = 0.32 \Omega$

- (ii) The line power factor needs to be improved to unity by installing capacitors at the input terminals of the induction motor. Calculate the per-phase capacitance required to obtain a line power factor of unity.

Solution (i) The equivalent circuit of the induction motor is utilized to solve the problem. First, calculate the equivalent input impedance of the induction motor. This is achieved by finding the equivalent impedance between the magnetizing and core-loss resistance, and then the combined impedance of this with the rotor impedance is found, which, when added to the stator impedance, gives the equivalent impedance of the induction machine.

$$\text{Magnetizing-branch equivalent impedance, } Z_o = \frac{jX_m R_c}{R_c + jX_m} = 5.47 + j49.39 \, \Omega$$

$$\text{Rotor impedance, } Z_r = \frac{R_r}{s} + jX_{lr} = \frac{0.12}{0.03447} + j0.32 \, \Omega$$

$$\text{Equivalent rotor and magnetizing impedance, } Z_{eq} = \frac{Z_o Z_r}{Z_o + Z_r} = 3.13 + j0.51 \, \Omega$$

$$\text{Motor equivalent impedance, } Z_{im} = R_s + jX_{ls} + Z_{eq} = 3.15 + j0.83 \, \Omega$$

$$\text{Phase input voltage, } V_{as} = \frac{V_{ll}}{\sqrt{3}} = \frac{2300}{\sqrt{3}} = 1327.9 \, \text{V}$$

$$\text{Stator current, } I_s = \frac{V_{as}}{Z_{im}} = 394.2 - j104.1 \, \text{A}$$

$$\text{Rotor current, } I_r = \frac{Z_{eq}}{Z_{im}} I_s = 393.87 - j78.07 \, \text{A}$$

$$\text{No-load current, } I_o = I_s - I_r = 0.33 - j26.03 \, \text{A}$$

$$\text{Core-loss current, } I_c = \frac{Z_o}{R_c} I_o = 2.85 - j0.28 \, \text{A}$$

Rotor angular speed,

$$\omega_m = (1 - s) \frac{\omega_s}{P/2} = (1 - s) \frac{2\pi f_s}{P/2} = 0.03746 \frac{2 * \pi * 60}{4/2} = 181.43 \, \text{rad/sec (mech)}$$

$$\text{Air-gap torque, } T_e = 3 \frac{P}{2} \frac{|I_r|^2 R_r}{s \omega_s} = 8220.1 \, \text{N}\cdot\text{m}$$

$$\text{Mechanical power, } P = \omega_m T_e = 1491.2 \, \text{kW}$$

$$\text{Shaft power output, } P_m = P_s - P_m = 1491.2 - 0 = 1491.2 \, \text{kW}$$

$$\text{Stator resistive losses, } P_{sc} = 3|I_s|^2 R_s = 3 * 407.74^2 * 0.2 = 9.975 \, \text{kW}$$

$$\text{Rotor resistive losses, } P_{rc} = 3|I_r|^2 R_r = 3 * 401.53^2 * 0.12 = 58.04 \, \text{kW}$$

$$\text{Core losses, } P_{co} = 3|I_c|^2 R_c = 2.865^2 * 451.2 = 11.11 \, (\text{kW})$$

$$\text{Input power, } P_i = P_m + P_{sc} + P_{rc} + P_{co} = 1491.2 + 9.975 + 58.04 + 11.11 = 1570.5 \, \text{kW}$$

$$\% \text{ Efficiency, } \eta = \frac{P_s}{P_i} 100 = \frac{1491.2}{1570.5} 100 = 94.96\%$$

(ii) The principle of power-factor improvement with capacitor installation at the machine stator terminals is based on the capacitor's drawing a leading reactive current from the supply to cancel the lagging reactive current drawn by the induction machine. In order for the line power factor to be unity, the reactive component of the line current must be zero. The reactive line current is the sum of the capacitor and induction machine reactive currents. Therefore, the capacitive reactive current (I_{cap}) has to be equal in magnitude but opposite in

direction to the machine lagging reactive current, but the machine reactive current is the imaginary part of the stator current and is given by

$$I_{\text{cap}} + \text{imag}(I_s) = 0$$

$$\text{Hence, } I_{\text{cap}} = -\text{imag}(I_s) = -(-j104.1) = j104.1 \text{ A}$$

This current is controlled by the capacitors installed at the input, and therefore the capacitor required is

$$C = \frac{I_{\text{cap}}}{j\omega_s V_{\text{as}}} = \frac{j104.1}{j377 * 1327.9} = 20.792 \mu\text{F}$$

5.5 STEADY-STATE PERFORMANCE

A flowchart for the evaluation of the steady-state performance of the motor is given in Figure 5.5. The relevant equations for use are from (5.22) to (5.33). A set of sample torque-vs.-slip characteristics for constant input voltage is shown in Figures 5.6(a) and (b).

The torque-vs.-slip characteristics are shown for slip varying from 0 to 1. The slip is chosen in place of rotor speed because it is nondimensional and so is applicable to any motor frequency. Near the synchronous speed, i.e., at low slips, the torque is linear and is proportional to slip; beyond the maximum torque (also known as breakdown torque), the torque is approximately inversely proportional to slip, as is seen from the Figure 5.6(a). At standstill, the slip equals unity, and, at this operating point, the torque produced is known as standstill torque. To accelerate a load, this standstill torque has to be greater than the load torque. It is preferable to operate near low slips to have higher efficiency. This is due to the fact that the rotor copper losses are directly proportional to slip and are equal to the slip power, and, hence, at low slips, the rotor copper losses are small.

The positive-slope region of the torque-slip characteristics provides stable operation. Consider the machine operating at 1 p.u. with a low slip, and let the load torque be increased to 1.5 p.u. The rotor slows down and thereby develops a larger slip, which increases the electromagnetic torque capable of meeting the load torque. The new steady state is reached at 1.5 p.u. torque after a transient period with oscillations in torque. If the operating point at 1 p.u. torque at a slip of 0.85 is considered, the load torque disturbance will lead to increasing slip, resulting in less and less torque generation, thereby diverging more and more from the new load torque leading to a final pullout of the machine and reaching standstill. As this discussion is addressed to the steady-state characteristics of the machine, the two regions are named statically stable and unstable, as indicated in Figure 5.6(a).

Figure 5.6(b) shows the torque vs. slip for a wide range of slip, from -2 to 2, with the stator supplied with rated voltages and frequency. Consider the motor as spinning in the direction opposite to that of a phase sequence *abc*. Assume that a set of stator voltages with a phase sequence *abc* is applied at supply frequency. This creates a stator flux linkage counter to the direction of rotor speed, resulting in a braking action. This also creates a slip greater than one: the rotor speed is negative with

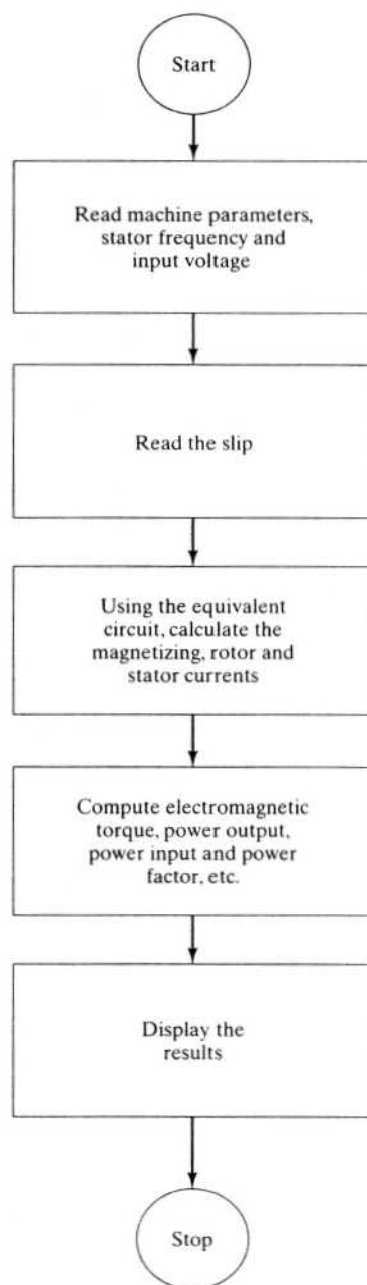


Figure 5.5 Flowchart for the computation of steady-state performance of the induction motor

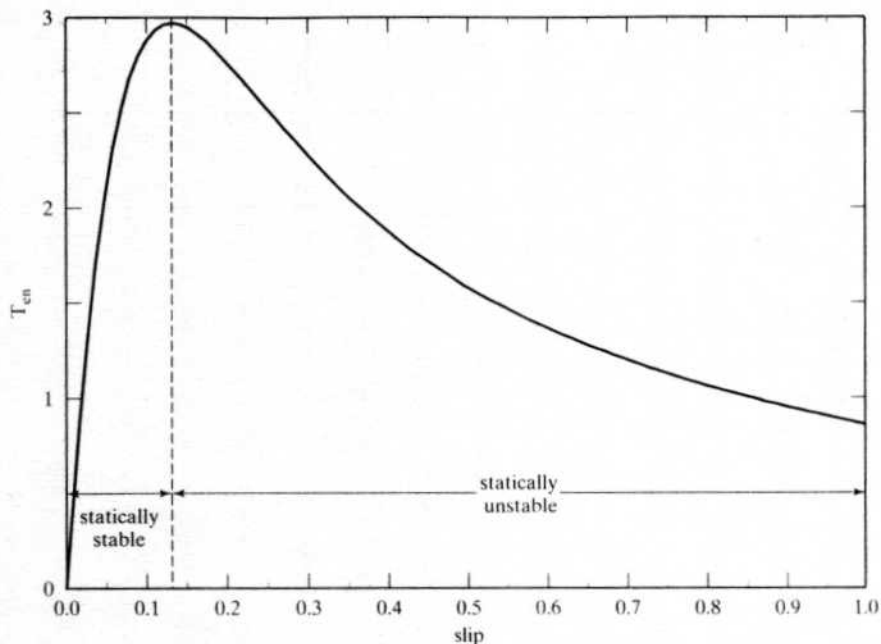


Figure 5.6 (a) Induction motor speed-torque characteristics

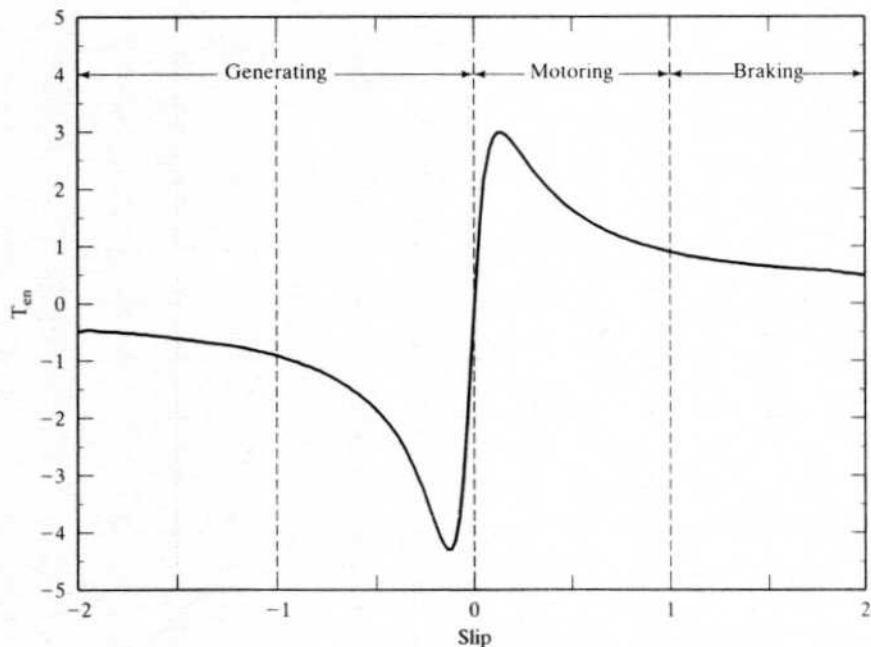


Figure 5.6 (b) Generation and braking characteristics of the induction motor

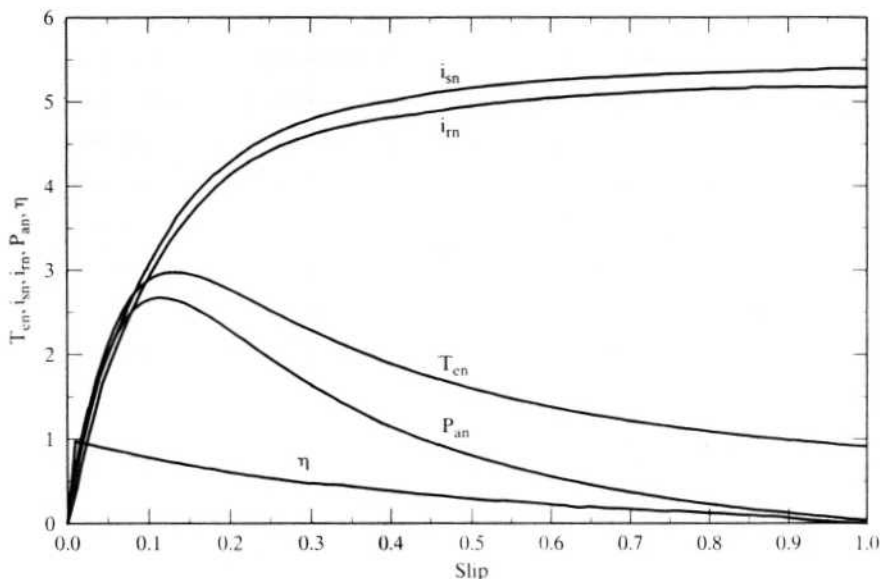


Figure 5.7 Performance characteristics of an induction motor

respect to synchronous speed. This braking action brings rotor speed to standstill in time. Consider the rotor electrical speed to be greater than the synchronous speed, resulting in a negative slip. A negative slip changes the generation of positive (motoring) to negative (generating) torque as the induced emf phase is reversed. Hence, for negative slip, the torque-vs.-slip characteristic is similar to the motoring characteristic discussed earlier, except that the breakdown torque is much higher with negative-slip operation. This is due to the fact that the mutual flux linkages are strengthened by the generator action of the induction machine. The reversal of rotor current reduces the motor impedance voltage drop, resulting in a boost of magnetizing current and hence in an increase of mutual flux linkages and torque.

The performance characteristics are shown in Figure 5.7 for rated voltage and frequency. The stator current at standstill on full voltage is 5.35 p.u. The efficiency is approximately equal to $(1 - s)$. Even though there is torque generated at standstill, the power output is zero (because of zero rotor speed). For variable-speed applications, the range of interest for the slip is 0 to 0.13 p.u., i.e., in the positive slope region of the torque-vs.-slip characteristics. It is seen that the efficiency is high in this region, as explained elsewhere.

Starting Torque: The starting torque is obtained from the torque expression by substituting for $s = 1$. Neglecting the magnetizing current, the electromagnetic torque at starting is approximately

$$T_e \cong \frac{3}{\omega_s} \cdot \frac{P}{2} \cdot \frac{V_{as}^2 \cdot R_r}{(R_s + R_r)^2 + (X_{ls} + X_{lr})^2} \quad (5.35)$$

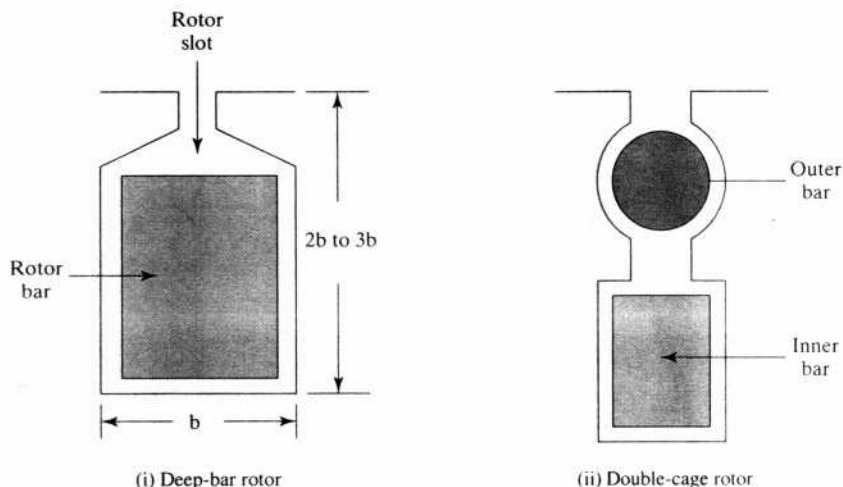


Figure 5.8 Deep-bar and double-cage rotors for the induction motor

This torque can be augmented for motors started directly with full line voltages (what are commonly termed *line start* motors), by increasing the rotor resistance. That increases the numerator proportionally and the denominator only slightly in equation (5.35), contributing to a higher torque. Rotor resistance is increased through the connection of external resistors in the slip-ring rotor. In the case of the squirrel-cage rotor, it is realized through either deep-bar or double-cage rotors, shown in Figure 5.8. Note that, at standstill, the rotor currents are at supply frequency, and therefore the rotor conductors have considerable skin effect, resulting in a higher resistance. Further, the larger slot height, 2 to 3 times the slot width, results in higher leakage flux, which prevents the flow of current in the inner part of the rotor bars, further decreasing the cross-sectional area of the rotor bar for current conduction and resulting in increased rotor resistance. When the machine is operating near synchronous speed, the rotor current frequency is very low, and hence the skin effect becomes negligible, resulting in an even flow of current in the rotor bar. That reduces the rotor resistance and hence provides an efficient operation of the motor in steady state. By similar reasoning, it could be seen that the outer bar becomes effective during starting and the inner bar predominates during steady-state operation in the double-cage rotor shown in Figure 5.8(ii). Note that the outer bars have smaller area of cross-section compared to the inner bars.

Induction Motor Classifications: The National Electrical Manufacturers Association (NEMA Class A, B, C, and D) has classified induction motors based on the torque-slip characteristics. Typical characteristics of a NEMA B and NEMA D induction motor are shown in Figure 5.9. Class A machine characteristic is similar to Class B machine. NEMA Class A and B are for general-purpose applications, such as fans and pumps. Class C has lower peak torque but has a higher starting torque than class B. NEMA Class C is for driving compressor pumps. NEMA Class D is to provide high starting torque and a wide statically stable speed range of operation, with the disadvantage of low efficiency compared to other types.

tation are described. There is a small but significant following among international engineers for this model. This fact necessitates an understanding of this model to follow their research work and patents using the space-phasor model. A number of worked examples are given to illustrate the key concepts and methods in the analysis of induction machines.

5.2 CONSTRUCTION AND PRINCIPLE OF OPERATION

5.2.1 Machine Construction

This section outlines some salient aspects of the machine construction but does not go into design aspects. Design details may be found in many of the standard texts, some of which are cited in the references.

Magnetic Part: The stator and rotor of the induction machine are made up of magnetic steel laminations of thickness varying from 0.0185 to 0.035 inch (0.47 to 0.875 mm), machine-punched with slots at the inner periphery for the stator and at the outer periphery for the rotor. These slots can be partially closed or fully open in the stator laminations to adjust the leakage inductances of the stator windings. In the rotor laminations, the slots can be partially open or fully closed. The rotor slots can be very deep compared to the width and might contain from two to five compartments to hold rotor bars of different shapes placed in them. In such a case, the rotor is known as a *deep bar* rotor. Depending on the number of parallel copper bars placed in a slot, the machine is referred to as a double-, triple-, or multiple-cage induction motor. The multiple-cage rotor is intended to maximize the electromagnetic torque during starting and to minimize the rotor copper losses during steady-state operation. The stator laminations are aligned and stacked in a fixture and pressed by heavy presses of capacity varying from 40 to 80 tons, to pack the laminations very closely and to remove the air gap between them. With these steps, the stator magnetic part of the machine is ready for insertion of windings.

The rotor windings are skewed by one half or a full slot pitch from one end to the other, to minimize or to completely cancel some of the time harmonics. To accommodate such skewing, the rotor laminations are assembled with a skew in a jig and fixture and then pressed to make the rotor magnetic block.

Stator and Rotor Windings: Consider a three-phase induction machine having three windings each on its stator and rotor. The phase windings are displaced in space from each other by 120 electrical degrees, where

$$\text{Electrical degrees} = (\text{Pairs of poles}) \times \text{Mechanical degrees} \quad (5.1)$$

The three-phase rotor windings are short-circuited either within the rotor or outside of the rotor, with or without external resistances connected to them. If the rotor windings are connected through slip rings mounted on the shaft and adjacent to the rotor of the induction motor to provide external access to the rotor windings, the machine is referred to as a *slip-ring induction* motor. Alternately, the rotor windings can be bars of copper or aluminum, with two end rings attached to short-circuit the

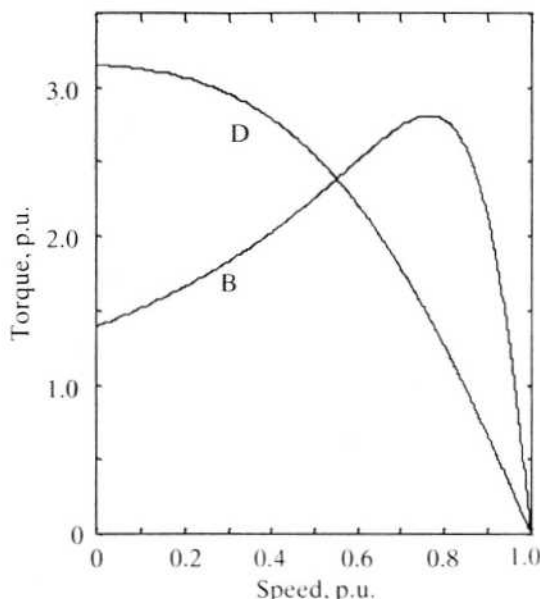


Figure 5.9 Typical speed–torque curves for squirrel-cage induction motors with NEMA Design classification B and D.

From the previous discussion on starting torques, it is seen that various types of motors are realized by means of different rotor slot and bar constructions.

5.6 MEASUREMENT OF MOTOR PARAMETERS

The measurement of motor parameters is based on the equivalent circuit of the induction motor. The various tests to measure the motor parameters are described, and the calculation of the parameters is considered.

5.6.1 Stator Resistance

With the rotor at standstill, the stator phase resistance is measured by applying a dc voltage and the resulting current. While this procedure gives only the dc resistance at a certain temperature, the ac resistance has to be calculated by considering the wire size, the stator frequency, and the operating temperature.

5.6.2 No-Load Test

The induction motor is driven at synchronous speed by another motor, preferably a dc motor. Then the stator is energized by applying rated voltage at rated frequency. The input power per phase, P_1 , is measured. The corresponding equivalent circuit for this condition is shown in Figure 5.10. The slip is zero, so the rotor circuit is open. For the present, the stator impedance can be neglected; it is small compared to the

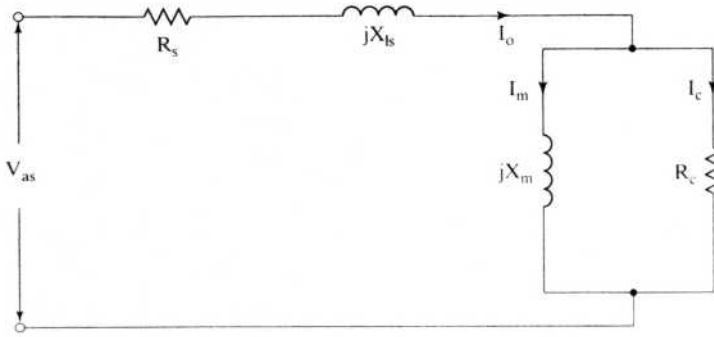


Figure 5.10 Equivalent circuit of the induction motor at no load

effective impedance of the magnetizing and core-loss branches. The mutual inductance and the core-loss resistance are calculated as follows:

The no-load power factor is given by

$$\cos \phi_o = \frac{P_1}{V_{as} I_o} \quad (5.36)$$

from which the magnetizing current is calculated as

$$I_m = I_o \sin \phi_o \quad (5.37)$$

and the core-loss current is given by

$$I_c = I_o \cos \phi_o \quad (5.38)$$

The magnetizing inductance is computed from

$$L_m = \frac{V_{as}}{2\pi f_s I_m} \quad (5.39)$$

and the core-loss resistance is given by

$$R_c = \frac{V_{as}}{I_c} \quad (5.40)$$

For most applications, such an approximate procedure for the evaluation of L_m and R_c is sufficient.

5.6.3 Locked-Rotor Test

The rotor of the induction motor is locked to keep it at standstill and a set of low three-phase voltages is applied to circulate rated stator currents. The input power per phase, P_{sc} , is measured along with the input voltage and stator current. The slip is unity for the locked-rotor condition and hence the circuit resembles that of a secondary-shortened transformer. The equivalent circuit for this situation is shown in Figure 5.11. The magnetizing branch impedance is very high compared to the rotor impedance, $(R_r + jX_{lr})$, and hence the approximate effective impedance is only the smaller of the two, which is the rotor impedance in this case. The parameters are calculated as follows.

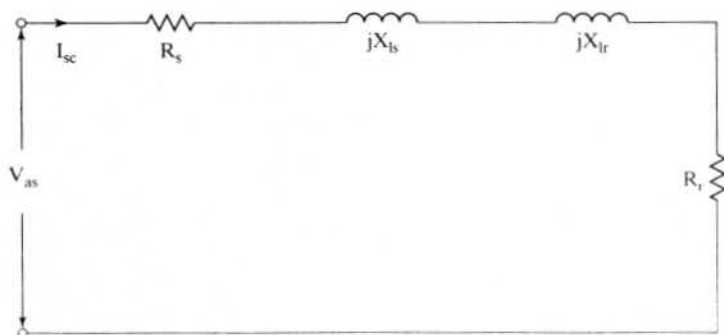


Figure 5.11 Equivalent circuit of the induction motor at standstill

The short-circuit power factor obtained from the equivalent circuit is

$$\cos \phi_{sc} = \frac{P_{sc}}{V_{sc} I_{sc}} \quad (5.41)$$

and the short-circuit impedance is given by

$$Z_{sc} = \frac{V_{sc}}{I_{sc}} \quad (5.42)$$

from which the rotor resistance and total leakage reactance are computed as

$$R_r = Z_{sc} \cos \phi_{sc} - R_s \quad (5.43)$$

$$X_{eq} = Z_{sc} \sin \phi_{sc} \quad (5.44)$$

where the total leakage reactance per phase, X_{eq} , is the sum of the stator and referred-rotor leakage reactances, given as

$$X_{eq} = X_{ls} + X_{lr} \quad (5.45)$$

It is usual to assume that the stator leakage inductance is equal to the rotor leakage inductance. For precise calculation, it is a good practice to consult the manufacturer's data sheet. Some rules of thumb are available as to their relationship for various types of motors.

Example 5.2

The no-load and locked-rotor test results for a three-phase, star-connected, 60-Hz, 2000-hp induction machine with a stator phase resistance of $0.02 \, \Omega$ are as follows:

Test	Input line to line voltage, V	Line current, A	Three-phase input power, kW
No load	2300	26.55	11.617
Locked rotor	462.68	407.75	319.22

Find the machine equivalent-circuit parameters.

Solution (i) From the no-load test results:

Input power, $P_1 = 11.617/3 = 3.872$ kW/phase

$$\text{Power factor, } \cos \phi_o = \frac{P_1}{V_{as} I_o} = \frac{11.617 \times 10^3}{\sqrt{3} \times 2300 \times 26.55} = 0.1098$$

Magnetizing current, $I_m = I_o \sin \phi_o = 26.39$ (A)

Core-loss branch current, $I_c = I_o \cos \phi_o = 2.9216$ (A)

Neglecting stator impedance, the following are calculated:

$$\text{Magnetizing inductance, } L_m = \frac{V_{as}}{2\pi f_s I_m} = \frac{2300 / \sqrt{3}}{2\pi \times 60 \times 26.39} = 0.1335 \text{ H}$$

$$\text{Core-loss resistance, } R_c = \frac{V_{as}}{I_c} = \frac{2300 / \sqrt{3}}{2.916} = 455.37 \text{ } (\Omega)$$

(ii) From the locked-rotor test results:

$$\text{Power factor, } \cos \phi_{sc} = \frac{P_{sc}}{V_{sc} I_{sc}} = \frac{(319.22 / 3) \times 10^3}{(462.68 / \sqrt{3}) \times 407.75} = 0.2137$$

$$\text{Phase angle, } \phi_{sc} = 1.355 \text{ rad}$$

$$\text{Locked-rotor impedance, } Z_{sc} = \frac{V_{sc}}{I_{sc}} = \frac{462.68 / \sqrt{3}}{407.75} = \frac{267.13}{407.75} = 0.6551 \Omega$$

$$\text{Stator-referred rotor resistance per phase, } R_r = Z_{sc} \cos \phi_{sc} - R_s = 0.12 \Omega$$

$$\text{Total leakage reactance, } X_{eq} = Z_{sc} \sin \phi_{sc} = 0.64 \Omega$$

The per-phase stator and rotor leakage reactances are half of the total leakage reactance; hence,

$$X_{ls} = X_{lr} = 0.32 \Omega$$

and the leakage inductances are

$$L_{ls} = L_{lr} = 0.8488 \text{ mH}$$

(iii) Further refinement of the equivalent-circuit parameters: The stator impedance was neglected in the calculation of magnetizing inductance and core-loss resistance because, at that time, the stator leakage inductance was not available. Now, having obtained it from the locked rotor test, we can use the no-load equivalent circuit to recalculate the magnetizing inductance and core-loss resistance; their final values are obtained as 0.1326 H and 451.2 Ω , respectively.

5.7 DYNAMIC MODELING OF INDUCTION MACHINES

The steady-state model and equivalent circuit developed in earlier sections are useful for studying the performance of the machine in steady state. This implies that all electrical transients are neglected during load changes and stator frequency variations. Such variations arise in applications involving variable-speed drives. The variable-speed drives are converter-fed from finite sources, unlike the utility

sources, due to the limitations of the switch ratings and filter sizes. This results in their incapability to supply large transient power. Hence, we need to evaluate the dynamics of converter-fed variable-speed drives to assess the adequacy of the converter switches and the converters for a given motor and their interaction to determine the excursions of currents and torque in the converter and motor. The dynamic model considers the instantaneous effects of varying voltages/currents, stator frequency, and torque disturbance. The dynamic model of the induction motor is derived by using a two-phase motor in direct and quadrature axes. This approach is desirable because of the conceptual simplicity obtained with two sets of windings, one on the stator and the other on the rotor. The equivalence between the three-phase and two-phase machine models is derived from simple observation, and this approach is suitable for extending it to model an n -phase machine by means of a two-phase machine. The concept of power invariance is introduced: the power must be equal in the three-phase machine and its equivalent two-phase model. The required transformation in voltages, currents, or flux linkages is derived in a generalized way. The reference frames are chosen to be arbitrary, and particular cases, such as stationary, rotor, and synchronous reference frames, are simple instances of the general case. Derivations for electromagnetic torque involving the currents and flux linkages are given. The differential equations describing the induction motor are nonlinear. For stability and controller design studies, it is important to linearize the machine equations around a steady-state operating point to obtain small-signal equations. Such a small-signal model is derived in this chapter. Algorithms and flow-chart are given to compute the eigenvalues, transfer functions, and frequency responses of the induction motor. Various stages in the dynamic modeling of the three-phase induction machine are shown in Figure 5.12. For the sake of completeness, the space-phasor model is derived from the dynamic model in direct and quadrature axes. The space-phasor model powerfully evokes the similarity and equivalence between the induction machines and dc machines from the modeling and control points of view. Examples are included in this section to illustrate the essential concepts.

5.7.1 Real-Time Model of a Two-Phase Induction Machine

The following assumptions are made to derive the dynamic model:

- (i) uniform air gap;
- (ii) balanced rotor and stator windings, with sinusoidally distributed mmf;
- (iii) inductance vs. rotor position is sinusoidal; and
- (iv) saturation and parameter changes are neglected.

A two-phase induction machine with stator and rotor windings is shown in Figure 5.13. The windings are displaced in space by 90 electrical degrees, and the rotor winding, α , is at an angle θ_r from the stator d axis winding. It is assumed that the d axis is leading the q axis for clockwise direction of rotation of the rotor. If the clockwise phase sequence is dq , the rotating magnetic field will be revolving at the angular speed of the supply frequency but counter to the phase sequence of the stator supply.

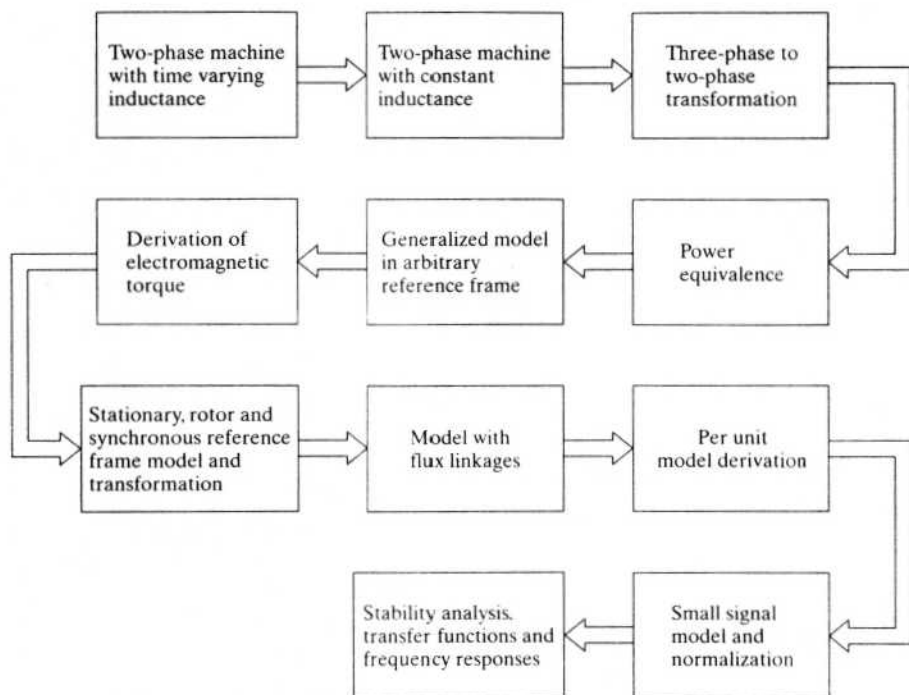


Figure 5.12 Dynamic modeling of the three-phase induction motor

Therefore, the rotor is pulled in the direction of the rotating magnetic field—that is, counter clockwise, in this case. The currents and voltages of the stator and rotor windings are marked in Figure 5.13. The number of turns per phase in the stator and rotor, respectively are T_1 and T_2 . A pair of poles is assumed for this figure, but it is applicable with slight modification for any number of pairs of poles if it is drawn in terms of electrical degrees. Note that θ_r is the electrical rotor position at any instant, obtained by multiplying the mechanical rotor position by pairs of electrical poles. The terminal voltages of the stator and rotor windings can be expressed as the sum of the voltage drops in resistances and rates of change of flux linkages, which are the products of currents and inductances. The equations are as follows:

$$v_{qs} = R_q i_{qs} + p(L_{qq} i_{qs}) + p(L_{qd} i_{ds}) + p(L_{q\alpha} i_{\alpha}) + p(L_{q\beta} i_{\beta}) \quad (5.46)$$

$$v_{ds} = p(L_{dq} i_{qs}) + R_d i_{ds} + p(L_{dd} i_{ds}) + p(L_{d\alpha} i_{\alpha}) + p(L_{d\beta} i_{\beta}) \quad (5.47)$$

$$v_{\alpha} = p(L_{\alpha q} i_{qs}) + p(L_{\alpha d} i_{ds}) + R_{\alpha} i_{\alpha} + p(L_{\alpha\alpha} i_{\alpha}) + p(L_{\alpha\beta} i_{\beta}) \quad (5.48)$$

$$v_{\beta} = p(L_{\beta q} i_{qs}) + p(L_{\beta d} i_{ds}) + p(L_{\beta\alpha} i_{\alpha}) + R_{\beta} i_{\beta} + p(L_{\beta\beta} i_{\beta}) \quad (5.49)$$

where p is the differential operator d/dt , and the various inductances are explained as follows. v_{qs} , v_{ds} , v_{α} , v_{β} are the terminal voltages of the stator q axis, d axis, and rotor α and β windings, respectively. i_{qs} and i_{ds} are the stator q axis and d axis currents, respectively. i_{α} and i_{β} are the rotor α and β windings currents, respectively. L_{qq} , L_{dd} , $L_{\alpha\alpha}$, and $L_{\beta\beta}$ are the stator q and d axes winding and rotor α and β winding self-inductances, respectively.

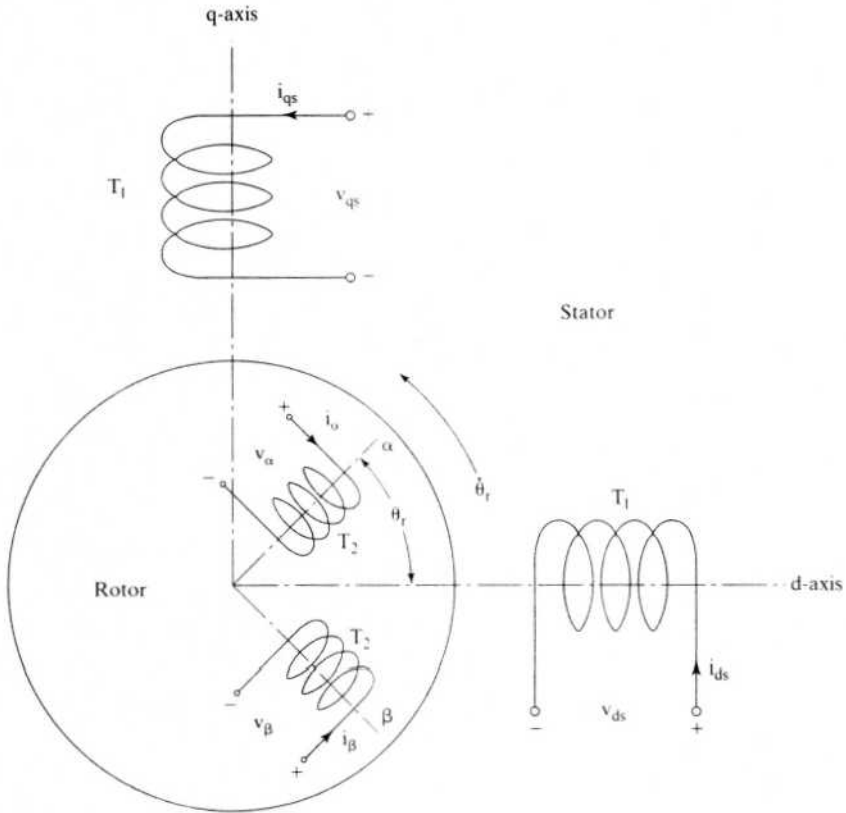


Figure 5.13 Stator and rotor windings of a two-phase induction motor

The mutual inductances between any two windings are denoted by L with two subscripts, the first subscript denoting the winding at which the emf is measured due to the current in the other winding, indicated by the second subscript. For example, L_{qd} is the mutual inductance between the q and d axes windings due to a current in the d axis winding. Under the assumption of uniform air gap, the self-inductances are independent of angular positions; hence, they are constants:

$$L_{\alpha\alpha} = L_{\beta\beta} = L_{rr} \quad (5.50)$$

$$L_{dd} = L_{qq} = L_s \quad (5.51)$$

The mutual inductances between the stator windings and between the rotor windings are zero, because the flux set up by a current in one winding will not link with the other winding displaced in space by 90 degrees. This leads to the following simplifications.

$$L_{\alpha\beta} = L_{\beta\alpha} = 0 \quad (5.52)$$

$$L_{dq} = L_{qd} = 0 \quad (5.53)$$

The mutual inductances between the stator and rotor windings are a function of the rotor position, θ_r , and they are assumed to be sinusoidal functions because of the assumption of sinusoidal mmf distribution in the windings. Symmetry in windings and construction causes the mutual inductances between one stator and one rotor winding to be the same whether they are viewed from the stator or the rotor:

$$L_{\alpha d} = L_{d\alpha} = L_{sr} \cos \theta_r \quad (5.54)$$

$$L_{\beta d} = L_{d\beta} = L_{sr} \sin \theta_r \quad (5.55)$$

$$L_{\alpha q} = L_{q\alpha} = L_{sr} \sin \theta_r \quad (5.56)$$

$$L_{\beta q} = L_{q\beta} = -L_{sr} \cos \theta_r \quad (5.57)$$

where L_{sr} is the peak value of the mutual inductance between a stator and a rotor winding. The last equation has a negative term, because a positive current in β winding produces a negative flux linkage in the q axis winding, and vice versa. Substitution of equations from (5.50) to (5.57) into equations from (5.46) to (5.49) results in a system of differential equations with time-varying inductances. The resulting equations are as follows:

$$v_{qs} = (R_s + L_s p) i_{qs} + L_{sr} p(i_{\alpha} \sin \theta_r) - L_{sr} p(i_{\beta} \cos \theta_r) \quad (5.58)$$

$$v_{ds} = (R_s + L_s p) i_{ds} + L_{sr} p(i_{\alpha} \cos \theta_r) + L_{sr} p(i_{\beta} \sin \theta_r) \quad (5.59)$$

$$v_{\alpha} = L_{sr} p(i_{qs} \sin \theta_r) + L_{sr} p(i_{ds} \cos \theta_r) + (R_{rr} + L_{rr} p) i_{\alpha} \quad (5.60)$$

$$v_{\beta} = -L_{sr} p(i_{qs} \cos \theta_r) + L_{sr} p(i_{ds} \sin \theta_r) + (R_{rr} + L_{rr} p) i_{\beta} \quad (5.61)$$

where

$$R_s = R_q = R_d \quad (5.62)$$

$$R_{rr} = R_{\alpha} = R_{\beta} \quad (5.63)$$

The solution of these equations is time-consuming, because of their dependence on the product of the instantaneous rotor-position-dependent cosinusoidal functions and winding currents, and an elegant set of equations leading to a simple solution procedure is necessary. Transformations performing such a step are discussed subsequently.

5.7.2 Transformation to Obtain Constant Matrices

The transformation to obtain constant inductances is achieved by replacing the actual with a fictitious rotor on the q and d axes, as shown in Figure 5.14. In that process, the fictitious rotor will have the same number of turns for each phase as the actual rotor phase windings and should produce the same mmf. That leads to a cancellation of the number of turns on both sides of that equation, resulting in a relationship of the actual currents to fictitious rotor currents i_{qrr} and i_{drr} . Then the fictitious rotor currents i_{qrr} and i_{drr} are equal to the sum of the projections of i_{α} and i_{β} on the q and d axis, respectively, as given below:

$$\begin{bmatrix} i_{drr} \\ i_{qrr} \end{bmatrix} = \begin{bmatrix} \cos \theta_r & \sin \theta_r \\ \sin \theta_r & -\cos \theta_r \end{bmatrix} \begin{bmatrix} i_{\alpha} \\ i_{\beta} \end{bmatrix} \quad (5.64)$$

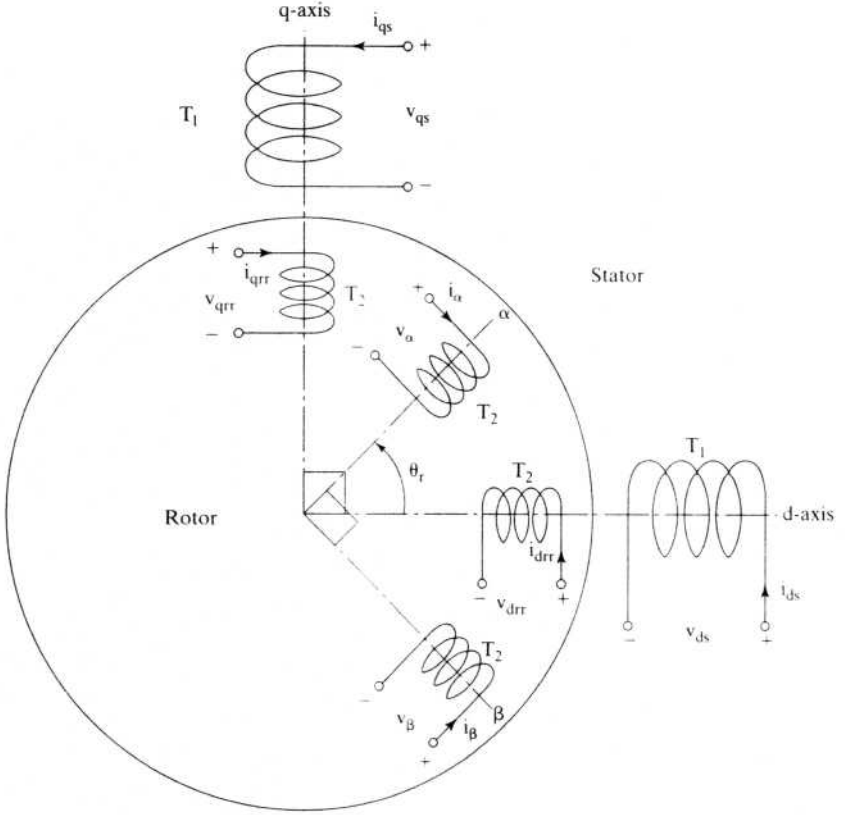


Figure 5.14 Transformation of actual to fictitious rotor variables

Equation (5.64) is written compactly as

$$\mathbf{i}_{dqrr} = [\mathbf{T}_{\alpha\beta}] \mathbf{i}_{\alpha\beta} \quad (5.65)$$

where

$$\mathbf{i}_{dqrr} = [i_{drr} \ i_{qrr}]^T \quad (5.66)$$

$$\mathbf{i}_{\alpha\beta} = [i_{\alpha} \ i_{\beta}]^T \quad (5.67)$$

and

$$\mathbf{T}_{\alpha\beta} = \begin{bmatrix} \cos \theta_r & \sin \theta_r \\ \sin \theta_r & -\cos \theta_r \end{bmatrix} \quad (5.68)$$

This transformation is valid for voltages, currents, and flux-linkages in a machine. That the transformation from α, β axes to d, q axes and vice versa is elegant is due to the fact that

$$\mathbf{T}_{\alpha\beta} = \mathbf{T}_{\alpha\beta}^{-1} \quad (5.69)$$

This matrix is both orthogonal and symmetric. Applying this transformation to the α and β rotor-winding currents and rotor voltages in equations (5.60) and (5.61), the following matrix equation is obtained:

$$\begin{bmatrix} v_{qs} \\ v_{ds} \\ v_{qrr} \\ v_{drr} \end{bmatrix} = \begin{bmatrix} R_s + L_s p & 0 & L_{sr} p & 0 \\ 0 & R_s + L_s p & 0 & L_{sr} p \\ L_{sr} p & -L_{sr} \dot{\theta}_r & R_{rr} + L_{rr} p & -L_{rr} \dot{\theta}_r \\ L_{sr} \dot{\theta}_r & L_{sr} p & L_{rr} \dot{\theta}_r & R_{rr} + L_{rr} p \end{bmatrix} \begin{bmatrix} i_{qs} \\ i_{ds} \\ i_{qrr} \\ i_{drr} \end{bmatrix} \quad (5.70)$$

where $\dot{\theta}_r$ is the time derivative of θ_r . The rotor equations need to be referred to the stator, as in the case of the transformer-equivalent circuit. This step removes the physical isolation and facilitates the corresponding stator and rotor d and q axes windings in becoming physically connected. The steps involved in referring these rotor parameters and variables to the stator are as follows:

$$R_r = a^2 R_{rr} \quad (5.71)$$

$$L_r = a^2 L_{rr} \quad (5.72)$$

$$i_{qr} = \frac{i_{qrr}}{a} \quad (5.73)$$

$$i_{dr} = \frac{i_{drr}}{a} \quad (5.74)$$

$$v_{qr} = a v_{qrr} \quad (5.75)$$

$$v_{dr} = a v_{drr} \quad (5.76)$$

where

$$a = \frac{\text{Stator effective turns per phase}}{\text{Rotor effective turns per phase}} = \frac{k_{w1} T_1}{k_{w2} T_2} \quad (5.77)$$

Note that the magnetizing and mutual inductances are,

$$L_m \propto T_1^2 \quad (5.78)$$

$$L_{sr} \propto T_1 T_2 \quad (5.79)$$

From equations (5.77) and (5.79), the magnetizing inductance of the stator is derived as

$$L_m = a L_{sr} \quad (5.80)$$

Substituting equations from (5.71) to (5.80) into (5.70), the machine equations referred to the stator are obtained as

$$\begin{bmatrix} v_{qs} \\ v_{ds} \\ v_{qr} \\ v_{dr} \end{bmatrix} = \begin{bmatrix} R_s + L_s p & 0 & L_m p & 0 \\ 0 & R_s + L_s p & 0 & L_m p \\ L_m p & -L_m \dot{\theta}_r & R_r + L_r p & -L_r \dot{\theta}_r \\ L_m \dot{\theta}_r & L_m p & L_r \dot{\theta}_r & R_r + L_r p \end{bmatrix} \begin{bmatrix} i_{qs} \\ i_{ds} \\ i_{qr} \\ i_{dr} \end{bmatrix} \quad (5.81)$$

bars on the rotor itself, thus making the rotor very compact. Such a construction is known as a *squirrel-cage induction motor*.

The numbers of slots in the stator and rotor are unequal to avoid harmonic crawling torques. The windings are distributed in the slots across the periphery of the stator and rotor. The windings can have different progressions, such as concentric or lap windings. In concentric windings, the windings are centered around slots within a pole pitch. In the case of lap windings, a coil is spread over a fixed pitch angle, say, 180 electrical degrees, and connected to the coil in the adjacent slot, and so on. With a coil pitch of 180 electrical degrees, the induced emf of the coil side under the north pole will be equal but opposite to the induced emf in the coil side under the south pole. The induced emf in the coil is equal to the sum of the induced emfs in both the coil sides, which will be twice the voltage induced in one coil side. The windings need not have a pitch of 180 electrical degrees and might have less than that, to eliminate some fixed number of harmonics. This case is known as short-chorded winding. Short chording reduces the resultant voltage, because the coils are not displaced by 180 degrees but by less, with the outcome that their phasor sum is less than their algebraic sum. Such an effect is included in the pitch factor, c_p .

The slots have a phase shift, both to allow for mechanical integrity and to control the short-chording angle, so the induced emf in the adjacent coils of a phase will have a phase displacement. When the emfs in these coils of a phase are summed up, the resultant induced emf is the sum of the phasor voltages induced in these coils. A reduction in the voltage results because of the phasor addition compared to their algebraic sum. The factor to account for this aspect is known as the distribution factor, c_d .

The resultant voltage in a phase winding is reduced both by the pitch and distribution factors. Therefore, the winding factor, k_w , which reflects the effectiveness of the winding, is given as

$$k_w = c_d c_p \quad (5.2)$$

Induced Emf: With this understanding, the induced emf in a stator phase winding is derived from the first principles. The mutual flux linkages are distributed sinusoidally. The induced emf is equal in magnitude to the rate of change of mutual flux linkages, which in turn is equal to the product of the effective number of turns in the winding and the mutual flux. From this, the induced emf in a phase winding is derived as

$$e_{as} = -\frac{d\lambda}{dt} = -\frac{d}{dt} \{(k_w T_1) \Phi_m \sin(2\pi f_s t)\} = -2\pi f_s k_w T_1 \Phi_m \cos(2\pi f_s t) \quad (5.3)$$

where f_s is the supply frequency in Hz, Φ_m is the peak value of the mutual flux, T_1 is the total number of turns in phase a , and k_{w1} is the stator winding factor. The rms value of the induced emf is given by

$$E_{as} = \frac{|e_{as}|}{\sqrt{2}} = 4.44 k_{w1} T_1 \Phi_m f_s \quad (5.4)$$

The expression for the induced emf in the rotor phase windings is very similar to equation (5.4) if appropriate winding factor, number of turns, and frequency for the

This equation is in the form where the voltage vector is equal to the product of the impedance matrix and current vector. Note that the impedance matrix has constant inductance terms and is no longer dependent on the rotor position. Some of the impedance matrix elements are dependent on the rotor speed, and only when they are constant, as in steady state, does the system of equations become linear. In the case of varying rotor speed and if its variation is dependent on the currents, then the system of equations becomes nonlinear. It is derived later that the electromagnetic torque, as a function of winding currents and rotor speed, is determined by the electromagnetic and load torques along with load parameters such as inertia and friction. In that case, it can be seen that the induction-machine system is nonlinear.

5.7.3 Three-Phase to Two-Phase Transformation

The model that has been developed so far is for a two-phase machine. Three-phase induction machines are common; two-phase machines are rarely used in industrial applications. A dynamic model for the three-phase induction machine can be derived from the two-phase machine if the equivalence between three and two phases is established. The equivalence is based on the equality of the mmf produced in the two-phase and three-phase windings and equal current magnitudes. Figure 5.15 shows the three-phase and two-phase windings. Assuming that each of the three-phase windings has T_l turns per phase and equal current magnitudes, the two-phase windings will have $3T_l/2$ turns per phase for mmf equality. The d and q axes mmfs are found by resolving the mmfs of the three phases along the d and q axes.

The common term, the number of turns in the winding, is canceled on either side of the equations, leaving the current equalities. The q axis is assumed to be lagging the a axis by θ_c . The relationship between dqo and abc currents is as follows:

$$\begin{bmatrix} i_{qs} \\ i_{ds} \\ i_o \end{bmatrix} = \frac{2}{3} \begin{bmatrix} \cos \theta_c & \cos \left(\theta_c - \frac{2\pi}{3} \right) & \cos \left(\theta_c + \frac{2\pi}{3} \right) \\ \sin \theta_c & \sin \left(\theta_c - \frac{2\pi}{3} \right) & \sin \left(\theta_c + \frac{2\pi}{3} \right) \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} i_{as} \\ i_{bs} \\ i_{cs} \end{bmatrix} \quad (5.82)$$

The current i_o represents the imbalances in the a , b , and c phase currents and can be recognized as the zero-sequence component of the current. Equation (5.79) can be expressed in a compact form by

$$i_{qdo} = [T_{abc}] i_{abc} \quad (5.83)$$

where

$$i_{qdo} = [i_{qs} \ i_{ds} \ i_o]^t \quad (5.84)$$

$$i_{abc} = [i_{as} \ i_{bs} \ i_{cs}]^t \quad (5.85)$$

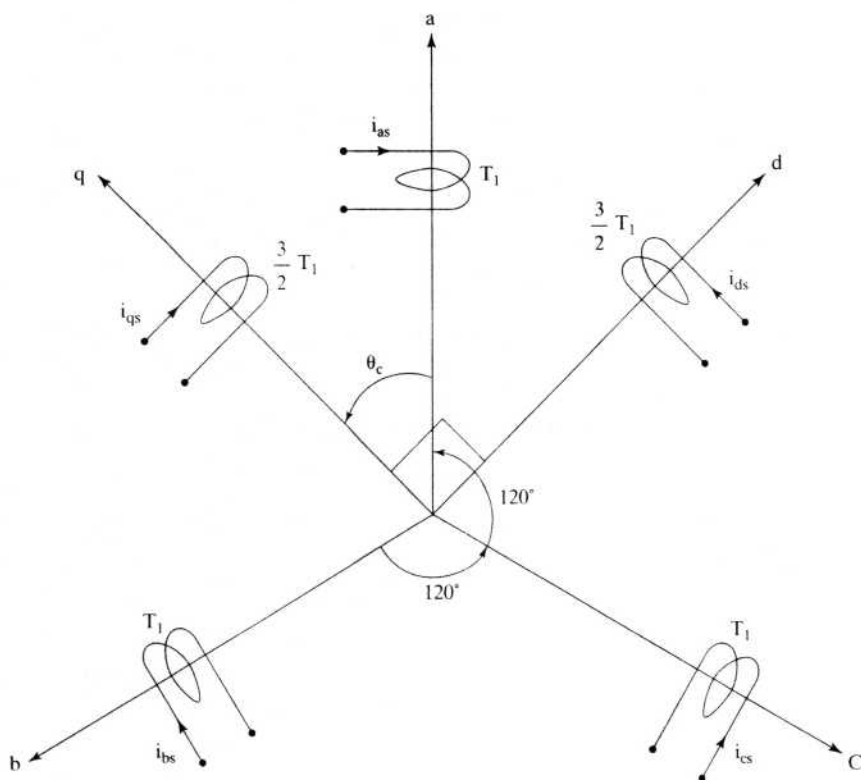


Figure 5.15 Two- and three-phase stator windings

and the transformation from abc to qdo variables is

$$[T_{abc}] = \frac{2}{3} \begin{bmatrix} \cos \theta_c & \cos \left(\theta_c - \frac{2\pi}{3} \right) & \cos \left(\theta_c + \frac{2\pi}{3} \right) \\ \sin \theta_c & \sin \left(\theta_c - \frac{2\pi}{3} \right) & \sin \left(\theta_c + \frac{2\pi}{3} \right) \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \quad (5.86)$$

The zero-sequence current, i_o , does not produce a resultant magnetic field. The transformation from two-phase currents to three-phase currents can be obtained as

$$i_{abc} = [T_{abc}]^{-1} i_{qdo} \quad (5.87)$$

where

$$[T_{abc}]^{-1} = \begin{bmatrix} \cos \theta_c & \sin \theta_c & 1 \\ \cos \left(\theta_c - \frac{2\pi}{3} \right) & \sin \left(\theta_c - \frac{2\pi}{3} \right) & 1 \\ \cos \left(\theta_c + \frac{2\pi}{3} \right) & \sin \left(\theta_c + \frac{2\pi}{3} \right) & 1 \end{bmatrix} \quad (5.88)$$

This transformation could also be thought of as a transformation from three (abc) axes to three new (qdo) axes; for uniqueness of the transformation from one set of axes to another set of axes, including unbalances in the abc variables requires three variables such as the dqo . The reason for this is that it is easy to convert from three abc variables to two qd variables if the abc variables have an inherent relationship among themselves, such as the equal-phase displacement and magnitude. Therefore, in such a case, there are only two independent variables in abc ; the third is a dependent variable, obtained as the negative sum of the other two variables. Hence a qd -to- abc transformation is unique under that circumstance. When the abc variables have no such inherent relationship, then there are three distinct and independent variables; hence, the third variable cannot be recovered from the knowledge of the other two variables only. It also means that they are not recoverable from two variables qd but require another variable, such as the zero-sequence component, to recover the abc variables from the dqo variables.

Under balanced conditions only, there are four system equations, as given in (5.81). Under unbalanced conditions, note that two more system equations, one for the stator zero-sequence voltage and the other for the rotor zero-sequence voltage, emerge. They are given as [8]

$$v_{os} = R_s + L_{ls} p i_{os} \quad (5.89)$$

$$v_{or} = R_r + L_{lr} p i_{or} \quad (5.90)$$

where in the variables the first subscript, o , denotes the zero-sequence component and the second subscript denotes the stator and rotor by s and r , respectively. They could be derived from the stator and rotor inductance matrices in the abc frames, then converted into dqo frames by using the transformation derived above. It is interesting to observe that only leakage inductances and phase resistances influence the zero-sequence voltages and currents, unlike in the dq component variables, which are influenced by the self and mutual inductances and phase resistances.

It is usual to align the q axis with the phase a winding; this implies that the qd frames are fixed to the stator. The model is known as Stanley's model or the *stator reference frames* model. In that case, $\theta_c = 0$, and the transformation from abc to dqo variables is given as

$$T_{abc}^s = \frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & -\frac{\sqrt{3}}{2} & \frac{\sqrt{3}}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \quad (5.91)$$

In a balanced three-phase machine, the sum of the three-phase currents is zero and is given as

$$i_{as} + i_{bs} + i_{cs} = 0 \quad (5.92)$$

leading to a zero-sequence current of zero value:

$$i_o = \frac{1}{3}(i_{as} + i_{bs} + i_{cs}) = 0 \quad (5.93)$$

With equations (5.86) and (5.88), the equivalence between the two-phase and three-phase induction machines is established. It is instructive to know that the transformation derived is applicable to currents, voltages, and flux-linkages.

Example 5.3

An induction motor has the following parameters:

$$\begin{aligned} &5 \text{ hp, } 200 \text{ V, 3-phase, } 60 \text{ Hz, 4-pole, star-connected; } R_s = 0.277 \, \Omega \\ &R_r = 0.183 \, \Omega \quad L_m = 0.0538 \text{ H} \quad L_s = 0.0553 \text{ H} \quad L_r = .056 \text{ H} \\ &\text{Effective stator to rotor turns ratio, } a = 3 \end{aligned}$$

The motor is supplied with its rated and balanced voltages. Find the q and d axes steady-state voltages and currents and phase currents I_{qr} , I_{dr} , I_a , and I_b when the rotor is locked. Use the stator-reference-frames model of the induction machine.

Solution The applied phase voltages are as follows:

$$\begin{aligned} v_{as} &= \frac{200}{\sqrt{3}} \times \sqrt{2} \sin \omega_s t = 163.3 \sin \omega_s t \\ v_{bs} &= 163.3 \sin \left(\omega_s t - \frac{2\pi}{3} \right) \\ v_{cs} &= 163.3 \sin \left(\omega_s t + \frac{2\pi}{3} \right) \end{aligned}$$

The d and q axes voltages are

$$\begin{bmatrix} v_{qs} \\ v_{ds} \\ v_o \end{bmatrix} = T_{abc}^s \begin{bmatrix} v_{as} \\ v_{bs} \\ v_{cs} \end{bmatrix}$$

$$\text{Hence, } v_{qs} = \frac{2}{3} \left[v_{as} - \frac{1}{2}(v_{bs} + v_{cs}) \right]$$

For a balanced three-phase input,

$$v_{as} + v_{bs} + v_{cs} = 0$$

Substituting for v_{bs} and v_{cs} in terms of v_{as} yields

$$v_{qs} = \frac{2}{3} \left[\frac{3}{2} v_{as} \right] = v_{as}$$

Similarly,

$$v_{ds} = \frac{1}{\sqrt{3}}(v_{cs} - v_{bs})$$

and $v_o = 0$

$$v_{qs} = v_{as} = 163.3 \sin \omega_s t = 163.3 \angle 0^\circ = 163.3 \text{ V}$$

$$v_{ds} = \frac{1}{\sqrt{3}}(v_{cs} - v_{bs}) = 163.3 \cos \omega_s t = 163.3 \angle 90^\circ = j163.3 \text{ V}$$

The rotor is locked; hence,

$$\dot{\theta}_r = 0$$

For steady state evaluation,

$$p = j\omega_s = j2\pi f_s = j2\pi \cdot 60 = j377 \text{ rad/sec}$$

The system equations in steady state are

$$\begin{bmatrix} v_{qs} \\ v_{ds} \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} R_s + j\omega_s L_s & 0 & j\omega_s L_m & 0 \\ 0 & R_s + j\omega_s L_s & 0 & j\omega_s L_m \\ j\omega_s L_m & 0 & R_r + j\omega_s L_r & 0 \\ 0 & j\omega_s L_m & 0 & R_r + j\omega_s L_r \end{bmatrix} \begin{bmatrix} i_{qs} \\ i_{ds} \\ i_{qr} \\ i_{dr} \end{bmatrix}$$

Note that the rotor windings are short-circuited, and hence rotor voltages are zero. The numerical values for the parameters and variables are substituted to solve for the currents. The currents are

$$i_{qs} = 35.37 - j108.18 = 113.81 \angle -71.90^\circ$$

$$i_{ds} = 108.18 + j35.37 = 113.81 \angle 18.10^\circ$$

$$i_{qr} = -34.88 + j103.63 = 109.34 \angle 108.6^\circ$$

$$i_{dr} = -103.63 - j34.88 = 109.34 \angle -161.4^\circ$$

Note that the stator and rotor currents are displaced by 90° among themselves, as expected in a two-phase machine. The zero-sequence currents are zero, because zero-sequence voltages are nonexistent with balanced supply voltages.

The phase currents are

$$\begin{bmatrix} i_{as} \\ i_{bs} \\ i_{cs} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ -1/2 & -\sqrt{3}/2 & 1 \\ -1/2 & \sqrt{3}/2 & 1 \end{bmatrix} \begin{bmatrix} i_{qs} \\ i_{ds} \\ i_o \end{bmatrix} = \begin{bmatrix} 113.8 \angle -71.9^\circ \\ 113.8 \angle 168.1^\circ \\ 113.8 \angle 48.1^\circ \end{bmatrix}$$

The various rotor currents are

$$i_{qrr} = a i_{qr} = 328.02 \angle 108.6^\circ$$

$$i_{drr} = a i_{dr} = 328.02 \angle -161.4^\circ$$

The α and β currents, assuming $\theta_r = 0$, are

$$\begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} = \begin{bmatrix} \cos \theta_r & \sin \theta_r \\ \sin \theta_r & -\cos \theta_r \end{bmatrix} \begin{bmatrix} i_{drr} \\ i_{qrr} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} i_{drr} \\ i_{qrr} \end{bmatrix} = \begin{bmatrix} 328.02 \angle -161.4^\circ \\ -328.02 \angle 108.6^\circ \end{bmatrix}$$

Example 5.4

Derive the steady-state equivalent circuit from the dynamic equations of the induction motor.

Solution The steady-state equivalent circuit of the induction motor can be derived by substituting for the d and q axes voltages in the system equations:

$$v_{as} = V_m \sin \omega_s t$$

$$v_{bs} = V_m \sin \left(\omega_s t - \frac{2\pi}{3} \right)$$

$$v_{cs} = V_m \sin \left(\omega_s t + \frac{2\pi}{3} \right)$$

Hence,

$$\begin{bmatrix} v_{qs} \\ v_{ds} \\ v_o \end{bmatrix} = [T_{abc}] \begin{bmatrix} v_{as} \\ v_{bs} \\ v_{cs} \end{bmatrix} = \begin{bmatrix} V_m \sin \omega_s t \\ V_m \cos \omega_s t \\ 0 \end{bmatrix} = \begin{bmatrix} V_m \angle 0^\circ \\ V_m \angle 90^\circ \\ 0 \end{bmatrix}$$

In steady state,

$$p = j\omega_s$$

$$v_{qr} = v_{dr} = 0$$

Substituting these into the system equations yields

$$\begin{bmatrix} V_m \\ jV_m \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} R_s + j\omega_s L_s & 0 & j\omega_s L_m & 0 \\ 0 & R_s + j\omega_s L_s & 0 & j\omega_s L_m \\ j\omega_s L_m & -\omega_r L_m & R_r + j\omega_s L_r & -L_r \omega_r \\ \omega_r L_m & j\omega_s L_m & L_r \omega_r & R_r + j\omega_s L_r \end{bmatrix} \begin{bmatrix} i_{qs} \\ i_{ds} \\ i_{qr} \\ i_{dr} \end{bmatrix}$$

The input voltages are in quadrature, so the currents have to be in quadrature, because the system in steady state is linear, and they can be represented as

$$i_{ds} = j\dot{i}_{qs}$$

$$i_{dr} = j\dot{i}_{qr}$$

Substituting these equations into the above equation and considering only one stator and rotor equation with rms values yields

$$V_s = (R_s + j\omega_s L_s)I_s + j\omega_s L_m I_r$$

$$0 = jL_m (\omega_s - \omega_r)I_s + (R_r + j(\omega_s - \omega_r)L_r)I_r$$

where V_s is the stator rms voltage and I_s and I_r are the stator and rotor rms currents, respectively. Rearrange the rotor equation with the aid of

$$\omega_{sl} = \omega_s - \omega_r = s\omega_s$$

The rotor equation then is

$$0 = jL_m \omega_s I_s + \left(\frac{R_r}{s} + j\omega_s L_r \right) I_r$$

The rotor and stator equations, when combined, give the equivalent circuit, with an understanding that the sum of stator and rotor currents gives the magnetizing current. Note that the stator and stator-referred rotor self-inductances are equal to their magnetizing inductance and respective leakage inductances and are given as

$$L_s = L_m + L_{ls}$$

$$L_r = L_m + L_{lr}$$

5.7.4 Power Equivalence

The power input to the three-phase motor has to be equal to the power input to the two-phase machine to have meaningful interpretation in the modeling, analysis, and simulation. Such an identity is derived in this section. The three-phase instantaneous power input is

$$p_i = v_{abc}^t i_{abc} = v_{as} i_{as} + v_{bs} i_{bs} + v_{cs} i_{cs} \quad (5.94)$$

From equation (5.82), the *abc* phase currents and voltages are transformed into their equivalent *qd*-axes currents and voltages as

$$i_{abc} = [T_{abc}]^{-1} i_{qdo} \quad (5.95)$$

$$v_{abc} = [T_{abc}]^{-1} v_{qdo} \quad (5.96)$$

Substituting equations (5.95) and (5.96) into equation (5.94) gives the power input as

$$p_i = v_{qdo}^t ([T_{abc}]^{-1})^t [T_{abc}]^{-1} i_{qdo} \quad (5.97)$$

Expanding the right-hand side of the equation (5.97) gives the power input in *dqo* variables:

$$p_i = \frac{3}{2} ((v_{qs} i_{qs} + v_{ds} i_{ds}) + 2v_o i_o) \quad (5.98)$$

For a balanced three-phase machine, the zero-sequence current does not exist; hence, the power input is compactly represented by

$$p_i = \frac{3}{2} (v_{qs} i_{qs} + v_{ds} i_{ds}) \quad (5.99)$$

The model development has so far kept the *d* and *q* axes stationary with respect to the stator. These axes or frames are known as reference frames. A generalized treatment where the speed of the reference frames is arbitrary is derived in the next section. The input power given by equation (5.98) remains valid for all occasions, provided that the voltages and currents correspond to the frames under consideration.

5.7.5 Generalized Model in Arbitrary Reference Frames

Reference frames are very much like observer platforms, in that each of the platforms gives a unique view of the system at hand as well as a dramatic simplification of the system equations. For example, consider that, for the purposes of control, it is desirable to have the system variables as dc quantities, although the actual variables are sinusoidal. This could be accomplished by having a reference frame revolving at the same angular speed as that of the sinusoidal variable. As the reference frames are moving at an angular speed equal to the angular frequency of the sinusoidal supply, say, then the differential speed between them is reduced to zero, resulting in the sinusoid being perceived as a dc signal from the reference frames. Then, by moving to that plane, it becomes easier to develop a small-signal

equation out of a nonlinear equation, as the quiescent or operating point is described only by dc values; this then leads to the linearized system around an operating point. Now, it is easier to synthesize a compensator for the system by using standard linear control-system techniques. Likewise, the independent rotor field position determines the induced emf and affects the dynamic system equations of both the wound-rotor and permanent-magnet synchronous machines. Therefore, looking at the entire system from the rotor, i.e., rotating reference frames, the system inductance matrix becomes independent of rotor position, thus leading to the simplification and compactness of the system equations. Such advantages are many from using reference frames. Instead of deriving the transformations for each and every particular reference frame, it is advantageous to derive the general transformation for an arbitrary rotating reference frame. Then any particular reference frame model can be derived by substituting the appropriate frame speed and position in the generalized reference model.

Reference frames rotating at an arbitrary speed are hereafter called arbitrary reference frames. Other reference frames are particular cases of these arbitrary reference frames. The relationship between the stationary reference frames denoted by d and q axes and the arbitrary reference frames denoted by d^c and q^c axes are shown in Figure 5.16. From now on, the three-phase machine is assumed to have balanced windings and balanced inputs, thus making the zero-sequence components be zero and eliminating the zero-sequence equations from further consideration. Note that the zero-sequence equations have to be included only for unbalanced operation of the motor, a situation common with a fault in the machine or converter.

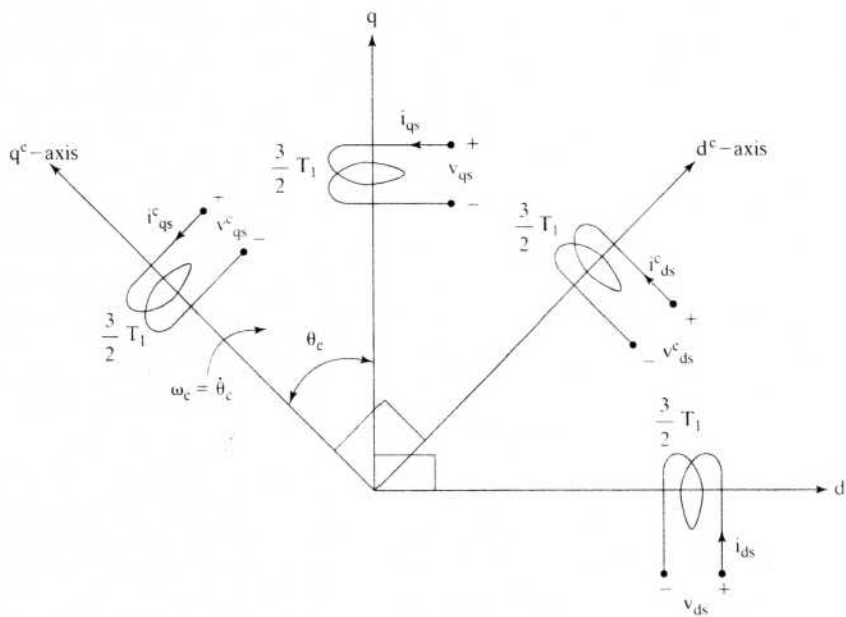


Figure 5.16 Stationary and arbitrary reference frames

Assuming that the windings have equal number of turns on both of the reference frames, the arbitrary reference frame currents are resolved on the d and q axes to find the currents in the stationary reference frames. The relationships between the currents are written as

$$\mathbf{i}_{qds} = [\mathbf{T}^c] \mathbf{i}_{qds}^c \quad (5.100)$$

where

$$\mathbf{i}_{qds} = \begin{bmatrix} i_{qs} & i_{ds} \end{bmatrix}^t \quad (5.101)$$

$$\mathbf{i}_{qds}^c = \begin{bmatrix} i_{qs}^c & i_{ds}^c \end{bmatrix}^t \quad (5.102)$$

and

$$\mathbf{T}^c = \begin{bmatrix} \cos \theta_c & \sin \theta_c \\ -\sin \theta_c & \cos \theta_c \end{bmatrix} \quad (5.103)$$

The speed of the arbitrary reference frames is

$$\dot{\theta}_c = \omega_c \quad (5.104)$$

Similarly, the fictitious rotor currents are transformed into arbitrary frames by using $\mathbf{T}^c$, and they are written as

$$\mathbf{i}_{qdr} = [\mathbf{T}^c] \mathbf{i}_{qdr}^c \quad (5.105)$$

where

$$\mathbf{i}_{qdr} = \begin{bmatrix} i_{qr} & i_{dr} \end{bmatrix}^t \quad (5.106)$$

$$\mathbf{i}_{qdr}^c = \begin{bmatrix} i_{qr}^c & i_{dr}^c \end{bmatrix}^t \quad (5.107)$$

Likewise, the voltage relationships are

$$\mathbf{v}_{qds} = [\mathbf{T}^c] \mathbf{v}_{qds}^c \quad (5.108)$$

$$\mathbf{v}_{qdr} = [\mathbf{T}^c] \mathbf{v}_{qdr}^c \quad (5.109)$$

where

$$\mathbf{v}_{qds} = \begin{bmatrix} v_{qs} & v_{ds} \end{bmatrix}^t \quad (5.110)$$

$$\mathbf{v}_{qds}^c = \begin{bmatrix} v_{qs}^c & v_{ds}^c \end{bmatrix}^t \quad (5.111)$$

$$\mathbf{v}_{qdr} = \begin{bmatrix} v_{qr} & v_{dr} \end{bmatrix}^t \quad (5.112)$$

$$\mathbf{v}_{qdr}^c = \begin{bmatrix} v_{qr}^c & v_{dr}^c \end{bmatrix}^t \quad (5.113)$$

By substituting equations from (5.100) to (5.113) into equation (5.81), the induction-motor model in arbitrary reference frames is obtained. It is given below for use in subsequent sections.

$$\begin{bmatrix} v_{qs}^c \\ v_{ds}^c \\ v_{qr}^c \\ v_{dr}^c \end{bmatrix} = \begin{bmatrix} R_s + L_s p & \omega_c L_s & L_m p & \omega_c L_m \\ -\omega_c L_s & R_s + L_s p & -\omega_c L_m & L_m p \\ L_m p & (\omega_c - \omega_r) L_m & R_r + L_r p & (\omega_c - \omega_r) L_r \\ -(\omega_c - \omega_r) L_m & L_m p & -(\omega_c - \omega_r) L_r & R_r + L_r p \end{bmatrix} \begin{bmatrix} i_{qs}^c \\ i_{ds}^c \\ i_{qr}^c \\ i_{dr}^c \end{bmatrix} \quad (5.114)$$

where

$$\omega_r = \dot{\theta}_r \quad (5.115)$$

ω_r is the rotor speed in electrical radians/sec. The relationship between the arbitrary reference frame variables and the a , b , and c variables is derived by using

$$i_{qds}^c = [T^c]^{-1} i_{qds} \quad (5.116)$$

By substituting from equation (5.91) for i_{qds} in terms of a , b , and c phase currents in the stator reference frames, the qdo currents in the arbitrary reference frames are obtained as

$$i_{qdo}^c = \begin{bmatrix} [T^c]^{-1} & 0 \\ 0 & 1 \end{bmatrix} [T_{abc}^s] [i_{abc}] = [T_{abc}] [i_{abc}] \quad (5.117)$$

where $[0]$ is a 1×2 null vector. Note that the zero-sequence currents remain unchanged in the arbitrary reference frames. This transformation is valid for currents, voltages, and flux-linkages for both the stator and the rotor. Particular cases of the reference frames are derived in a later section. The next section contains the derivation of the electromagnetic torque in terms of the current variables in the arbitrary reference frames.

5.7.6 Electromagnetic Torque

The electromagnetic torque is an important output variable that determines such mechanical dynamics of the machine as the rotor position and speed. Therefore, its importance cannot be overstated in any of the simulation studies. It is derived from the machine matrix equation by looking at the input power and its various components, such as resistive losses, mechanical power, rate of change of stored magnetic energy, and reference-frame power. Elementary reasoning leads to the fact that there cannot be a power component due to the introduction of reference frames. Similarly, the rate of change of stored magnetic energy must be zero in steady state. Hence, the output power is the difference between the input power and the resistive losses in steady state. Dynamically, the rate of change of stored magnetic energy need not be zero. Based on these observations, the derivation of the electromagnetic torque is made as follows.

The equation (5.114) can be written as

$$V = [R] i + [L] p i + [G] \omega_r i + [F] \omega_c i \quad (5.118)$$

where the vectors and matrices are identified by observation. Premultiplying the equation (5.118) by the transpose of the current vector gives the instantaneous input power as,

$$p_i = i^T V = i^T [R] i + i^T [L] p i + i^T [G] \omega_r i + i^T [F] \omega_c i \quad (5.119)$$

where the $[R]$ matrix consists of resistive elements, the $[L]$ matrix consists of the coefficients of the derivative operator p , the $[G]$ matrix has elements that are the coefficients of the electrical rotor speed ω_r , and $[F]$ is the frame matrix in terms of the coefficients of the reference frame speed, ω_c . The term $i^T [R] i$ gives stator and rotor resistive losses. The term $i^T [F] \omega_c i$ is the reference frame power, and upon expansion is found to be identically equal to zero, as it should be, because there cannot be a power

induced emf in the rotor are inserted into equation (5.4). More on this is deferred for the present.

Winding Method: Two types of winding methods are common in induction machines. They are known as *random-* and *form-wound* windings. They are briefly described in the following.

- (i) **Random-Wound Windings:** The coils are placed in the slots and separated from the magnetic steel with an insulation paper, such as a mica sheet. Each coil in a slot contains a number of circular but stranded enameled wires which are wound on a former. This type of winding is referred to as random-wound. They are used for low-voltage (<600 V) motors. The disadvantages of this method of winding the coil are: (i) The adjacent round wires in the worst case can be the first and the last turn in the coil. Because of this, the turn-to-turn voltage can be maximum in such a case and will equal the full coil voltage but not equal the sequential turn-to-turn voltage, which is only a small fraction of the full coil voltage. (ii) They are likely to have considerable air pockets between the round wires. These air pockets form capacitors between strands of wire. With the application of repetitive voltages at high frequency with high rate of change of voltages from the inverter, a discharge current flows into the air capacitor. This is known as partial discharge. These partial discharges cause insulation failures.

Random-wound machines are economical, have low losses, and hence have a higher efficiency and tend to run cooler. Random-wound machines use a semiclosed slot, which reduces the flux density in the teeth, resulting in lower core losses—as much as 20 to 30%. Some methods have very recently been suggested to reduce the partial-discharge possibility of random-wound machines. The methods recommend using (i) heavier insulation; (ii) a wind-in-place insertion method, making the wire placement sequential; (iii) extra strategically placed insulation within a phase of the motor; (iv) extra insulating sleeves on the turns nearer to the line leads.

- (ii) **Form-Wound Windings:** Higher-voltage (>600 V) induction machines are usually form-wound, meaning that each wire of rectangular cross section is placed in sequence with a strand of insulation in between them and then the bundle is wrapped with mica ground wall insulation, over which an armor covering is applied to keep all the wire arrangement firmly in place. The form-wound coils are placed in a fully open slot, as it is not possible to use the semi-closed slot for this arrangement, with the consequence that the machine will have higher core losses. This is because the tooth flux density increases: there is a sizable reduction of its cross sectional area compared to a semi-closed slotted tooth. On the other hand, form-wound windings have a much higher resistance to partial discharges, as their turn-to-turn voltage is minimal and a heavier insulation build between them also helps mitigate them. The disadvantage of this winding is that it is relatively expensive compared to random windings.

Rotor Construction: Two methods of rotor construction are used for induction machines. They are *fabrication* and *die casting* of the rotor. Fabricated rotor construction is possible for both aluminum and copper bar rotors, but aluminum

associated with a fictitious element introduced for the sake of simplifying the model and analysis. The term $i^1 [L] p i$ denotes the rate of change of stored magnetic energy. Therefore, what is left of the power component must be equal to the air gap power, given by the term $i^1 [G] \omega_r i$. From the fundamentals, it is known that the air gap power has to be associated with the rotor speed. The air gap power is the product of the mechanical rotor speed and air gap or electromagnetic torque. Hence air gap torque, T_e , is derived from the terms involving the rotor speed, ω_m in mechanical rad/sec, as

$$\omega_m T_e = P_a = i^1 [G] i \times \omega_r \quad (5.120)$$

Substituting for ω_r in terms of ω_m leads to electromagnetic torque as

$$T_e = \frac{P}{2} i^1 [G] i \quad (5.121)$$

By substituting for $[G]$ in equation (5.121) by observation from (5.114), the electromagnetic torque is obtained as

$$T_e = \frac{3}{2} \frac{P}{2} L_m (i_{qs}^{ic} i_{dr}^{ic} - i_{ds}^{ic} i_{qr}^{ic}) \quad (5.122)$$

The factor $\frac{3}{2}$ is introduced into the right-hand side of equation (5.122) from the power-equivalence condition between the three-phase and two-phase induction motors. The next section considers the frequently used models in various reference frames and their derivation from the generalized induction-motor model in arbitrary reference frames.

5.7.7 Derivation of Commonly Used Induction-Motor Models

Three particular cases of the generalized model of the induction motor in arbitrary reference frames are of general interest:

- (i) stator reference frames model;
- (ii) rotor reference frames model;
- (iii) synchronously rotating reference frames model.

Their derivations and transformation relationships are considered in this section.

5.7.7.1 Stator reference frames model. The speed of the reference frames is that of the stator, which is zero; hence,

$$\omega_e = 0 \quad (5.123)$$

is substituted into equation (5.114). The resulting model is

$$\begin{bmatrix} v_{qs} \\ v_{ds} \\ v_{qr} \\ v_{dr} \end{bmatrix} = \begin{bmatrix} R_s + L_s p & 0 & L_m p & 0 \\ 0 & R_s + L_s p & 0 & L_m p \\ L_m p & -\omega_r L_m & R_r + L_r p & -\omega_r L_r \\ \omega_r L_m & L_m p & \omega_r L_r & R_r + L_r p \end{bmatrix} \begin{bmatrix} i_{qs} \\ i_{ds} \\ i_{qr} \\ i_{dr} \end{bmatrix} \quad (5.124)$$

For convenience, the superscript c is omitted for the stator reference frames model hereafter. The torque equation is

$$T_e = \frac{3}{2} \frac{P}{2} L_m (i_{qs} i_{dr} - i_{ds} i_{qr}) \quad (5.125)$$

Note that equations (5.124) and (5.81) are identical. The transformation for variables is obtained by substituting $\theta_c = 0$ in $[T_{abc}]$ and will be the same as $[T_{abc}^s]$, defined in equation (5.91). This model is used when stator variables are required to be actual, i.e., the same as in the actual machine stator, and rotor variables can be fictitious. This model allows elegant simulation of stator-controlled induction-motor drives, such as phase-controlled and inverter-controlled induction-motor drives, because the input variables are well defined and could be used to find the stator q and d axes voltages through a set of simple algebraic equations, for a balanced polyphase supply input, given by:

$$v_{qs} = v_{as} \quad (5.126)$$

$$v_{ds} = \frac{(v_{cs} - v_{bs})}{\sqrt{3}} \quad (5.127)$$

Such algebraic relationships reduce the number of computations and thus lend themselves to real-time control applications in high-performance variable-speed drives requiring the computation of stator currents, stator flux linkages, and electromagnetic torque for both control and parameter adaptation.

5.7.7.2 Rotor reference frames model. The speed of the rotor reference frames is

$$\omega_c = \omega_r \quad (5.128)$$

and the angular position is

$$\theta_c = \theta_r \quad (5.129)$$

Substituting in the upper subscript r for rotor reference frames and equation (5.128) in the equation (5.114), the induction-motor model in rotor reference frames is obtained. The equations are given by

$$\begin{bmatrix} v_{qs}^r \\ v_{ds}^r \\ v_{qr}^r \\ v_{dr}^r \end{bmatrix} = \begin{bmatrix} R_s + L_s p & \omega_r L_s & L_m p & \omega_r L_m \\ -\omega_r L_s & R_s + L_s p & -\omega_r L_m & L_m p \\ L_m p & 0 & R_r + L_r p & 0 \\ 0 & L_m p & 0 & R_r + L_r p \end{bmatrix} \begin{bmatrix} i_{qs}^r \\ i_{ds}^r \\ i_{qr}^r \\ i_{dr}^r \end{bmatrix} \quad (5.130)$$

and the electromagnetic torque is

$$T_e = \frac{3}{2} \frac{P}{2} L_m (i_{qs}^r i_{dr}^r - i_{ds}^r i_{qr}^r) \quad (5.131)$$

The transformation from abc to dqo variables is obtained by substituting (5.129) into $[T_{abc}]$, defined in (5.86) as

$$[\mathbf{T}_{abc}^r] = \frac{2}{3} \begin{bmatrix} \cos \theta_r & \cos \left(\theta_r - \frac{2\pi}{3} \right) & \cos \left(\theta_r + \frac{2\pi}{3} \right) \\ \sin \theta_r & \sin \left(\theta_r - \frac{2\pi}{3} \right) & \sin \left(\theta_r + \frac{2\pi}{3} \right) \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \quad (5.132)$$

The rotor reference frames model is useful where the switching elements and power are controlled on the rotor side. Slip-power recovery scheme is one example where this model will find use in the simulation of the motor-drive system.

5.7.7.3 Synchronously rotating reference frames model. The speed of the reference frames is

$$\omega_c = \omega_s = \text{Stator supply angular frequency/rad/sec} \quad (5.133)$$

and the instantaneous angular position is

$$\theta_c = \theta_s = \omega_s t \quad (5.134)$$

By substituting (5.134) into (5.114), the induction-motor model in the synchronous reference frames is obtained. By using the superscript e to denote this electrical synchronous reference frame, the model is obtained as

$$\begin{bmatrix} v_{qs}^e \\ v_{ds}^e \\ v_{qr}^e \\ v_{dr}^e \end{bmatrix} = \begin{bmatrix} R_s + L_s p & \omega_s L_s & L_m p & \omega_s L_m \\ -\omega_s L_s & R_s + L_s p & -\omega_s L_m & L_m p \\ L_m p & (\omega_s - \omega_r) L_m & R_r + L_r p & (\omega_s - \omega_r) L_r \\ -(\omega_s - \omega_r) L_m & L_m p & -(\omega_s - \omega_r) L_r & R_r + L_r p \end{bmatrix} \begin{bmatrix} i_{qs}^e \\ i_{ds}^e \\ i_{qr}^e \\ i_{dr}^e \end{bmatrix} \quad (5.135)$$

The electromagnetic torque is,

$$T_e = \frac{3}{2} \frac{P}{2} L_m (i_{qs}^e i_{dr}^e - i_{ds}^e i_{qr}^e) \quad (\text{N}\cdot\text{m}) \quad (5.136)$$

The transformation from abc to dqo variables is found by substituting (5.134) into equation (5.86) and is given as

$$[\mathbf{T}_{abc}^e] = \frac{2}{3} \begin{bmatrix} \cos \theta_s & \cos \left(\theta_s - \frac{2\pi}{3} \right) & \cos \left(\theta_s + \frac{2\pi}{3} \right) \\ \sin \theta_s & \sin \left(\theta_s - \frac{2\pi}{3} \right) & \sin \left(\theta_s + \frac{2\pi}{3} \right) \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \quad (5.137)$$

It may be seen that the synchronous reference frames transform the sinusoidal inputs into dc signals. This model is useful where the variables in steady state need to be dc quantities, as in the development of small-signal equations. Some

high-performance control schemes use this model to estimate the control inputs; this led to a major breakthrough in induction-motor control, by decoupling the torque and flux channels for control in a manner similar to that for separately-excited dc motor drives. This is dealt with in detail in Chapter 8, on vector control schemes.

Example 5.5

Solve the problem given in Example 5.1 when its speed is 1400 rpm, using the synchronous reference frames model and rotating reference frames model.

Solution (i) Synchronous Reference Frames

$$\begin{bmatrix} v_{qs}^e \\ v_{ds}^e \\ v_o \end{bmatrix} = [T_{abc}^e] \begin{bmatrix} v_{as} \\ v_{bs} \\ v_{cs} \end{bmatrix}$$

where

$$v_{as} = V_m \sin \omega_s t$$

$$v_{bs} = V_m \sin\left(\omega_s t - \frac{2\pi}{3}\right)$$

$$v_{cs} = V_m \sin\left(\omega_s t + \frac{2\pi}{3}\right)$$

and

$$V_m = 163.3 \text{ V}$$

Substituting these, and solving for the d and q axes stator voltages in the synchronous reference frames, yields

$$\begin{bmatrix} v_{qs}^e \\ v_{ds}^e \\ v_o \end{bmatrix} = [T_{abc}^e] \begin{bmatrix} v_{as} \\ v_{bs} \\ v_{cs} \end{bmatrix} = \begin{bmatrix} 0 \\ V_m \\ 0 \end{bmatrix}$$

The d and q axes stator voltages are dc quantities; hence, the responses will be dc quantities, too, because the system is linear. Hence,

$$p i_{qs}^e = p i_{ds}^e = p i_{qr}^e = p i_{dr}^e = 0$$

Substituting these into the system equations gives

$$\begin{bmatrix} 0 \\ V_m \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} R_s & \omega_s L_s & 0 & \omega_s L_m \\ -\omega_s L_s & R_s & -\omega_s L_m & 0 \\ 0 & \omega_{sl} L_m & R_r & \omega_{sl} L_r \\ -\omega_{sl} L_m & 0 & -\omega_{sl} L_r & R_r \end{bmatrix} \begin{bmatrix} i_{qs}^e \\ i_{ds}^e \\ i_{qr}^e \\ i_{dr}^e \end{bmatrix}$$

where

$$\omega_{sl} = \omega_s - \omega_r = \text{slip speed}$$

The currents are solved by inverting the impedance matrix and premultiplying with the input-voltage vector. The currents in the synchronous frame are

$$\begin{bmatrix} i_{qs}^e \\ i_{ds}^e \\ i_{qr}^e \\ i_{dr}^e \end{bmatrix} = \begin{bmatrix} -9.27 \text{ A} \\ 14.35 \text{ A} \\ -1.69 \text{ A} \\ -14.60 \text{ A} \end{bmatrix}$$

The electromagnetic torque is given by

$$T_e = \frac{3}{2} \frac{P}{2} L_m (i_{qs}^e i_{dr}^e - i_{ds}^e i_{qr}^e) (\text{N}\cdot\text{m}) = 25.758 \text{ N}\cdot\text{m}$$

The actual phase currents are obtained by the following procedure:

$$i_{abc} = [T_{abc}]^{-1} i_{qdo}^e$$

Hence,

$$i_{as} = 17.08 \sin(\omega_s t - 0.57)$$

$$i_{bs} = 17.08 \sin(\omega_s t - 2.67)$$

$$i_{cs} = 17.08 \sin(\omega_s t + 1.52)$$

Note that the three-phase currents are balanced and that the power-factor angle is 0.57 radians (32.9°) lagging.

(ii) Rotor Reference Frames

$$\begin{bmatrix} v_{qs}^r \\ v_{ds}^r \\ v_o \end{bmatrix} = [T_{abc}^r] \begin{bmatrix} v_{as} \\ v_{bs} \\ v_{cs} \end{bmatrix}$$

Substituting for the transformation from equation (5.118) gives

$$\begin{bmatrix} v_{qs}^r \\ v_{ds}^r \\ v_o \end{bmatrix} = \begin{bmatrix} V_m \sin \omega_{sl} t \\ V_m \cos \omega_{sl} t \\ 0 \end{bmatrix} = \begin{bmatrix} 0 - jV_m \\ V_m + j0 \\ 0 \end{bmatrix}$$

The stator voltages appear at slip frequency in rotor reference frames; hence, the currents are at slip frequency in steady state. By substituting for $p = j\omega_{sl}$ in the system equations, the following is obtained:

$$\begin{bmatrix} -jV_m \\ V_m \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} R_s + j\omega_{sl}L_s & \omega_r L_s & j\omega_{sl}L_m & \omega_r L_m \\ -\omega_r L_s & R_s + j\omega_{sl}L_s & -\omega_r L_m & j\omega_{sl}L_m \\ j\omega_{sl}L_m & 0 & R_r + j\omega_{sl}L_r & 0 \\ 0 & j\omega_{sl}L_m & 0 & R_r + j\omega_{sl}L_r \end{bmatrix} \begin{bmatrix} i_{qs}^r \\ i_{ds}^r \\ i_{qr}^r \\ i_{dr}^r \end{bmatrix}$$

from which the rotor reference frame currents are

$$\begin{bmatrix} i_{qs}^r \\ i_{ds}^r \\ i_{qr}^r \\ i_{dr}^r \end{bmatrix} = \begin{bmatrix} 17.08 \sin(\omega_{sl} t - 0.57) \\ 17.08 \cos(\omega_{sl} t - 0.57) \\ 14.69 \sin(\omega_{sl} t - 3.03) \\ 14.69 \cos(\omega_{sl} t - 3.03) \end{bmatrix}$$

from which, by using the inverse transformation $[T_{abc}^r]^{-1}$, the actual phase stator currents can be evaluated. It can be seen that the stator q and d axes currents are displaced from each